Recovering LLM-Persona Accuracies from Unlabeled Votes

An algebraic NTQR evaluation study, and why disagreement is not enough

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1 Abstract

Algebraic (NTQR) evaluation infers how accurate a group of noisy classifiers was on a finite test **using only their responses — no answer key**. We test this end to end on real large language models. Three trader "personas" (optimistic, neutral, pessimistic), instantiated as system prompts, each make a binary `bullish`/`bearish` call on the same 64 market scenarios; we run the identical trio through six locally-hosted models via Ollama. For each model we recover per-persona, per-label accuracy with `ErrorIndependentEvaluation` (unsupervised) and score it against the authored ground truth (supervised), which is used only as a check.

On the five models whose three judges all varied (`mistral:latest`, `gemma4:latest`, `gemma3:4b`, `gemma2:2b`, `granite4.1:3b`), the unsupervised algebra recovered persona accuracies to a mean absolute error of **0.012**, within the 0.102 sampling-noise floor across all six per-label accuracy terms, with no labels – including a persona's genuinely poor bullish accuracy of **0.57**, recovered as **0.59**. The other model collapsed at least one persona into a **constant classifier** (a judge that voted one way on all 64 scenarios), which makes the error-independent algebra unsolvable.

The central, non-obvious result: **inter-judge disagreement does not imply evaluability.** Aggregate disagreement separated this run only because the unevaluable model(s) collapsed to 0.00; the five evaluable models spanned 0.03–0.23. What gates evaluation is a *per-judge* condition — every judge must vary (and answer) — not an *ensemble* one. We formalize this as a label-free **evaluability diagnostic** (a judge whose modal-vote fraction reaches 1.0 is a constant classifier; an unparseable vote is an abstention) that predicted exactly which models would be evaluable, before any solve and without ground truth. This is a concrete instance of the safety property the NTQR logic promises: it warns you when an ensemble is not good enough to be evaluated. A scenario bootstrap puts a 95% CI of [0.000, 0.038] on the recovery MAE (well inside the 0.102 noise floor), and a deterministic synthetic study generalizes the recovery beyond the finite set of real evaluable models — error falls like $1/\sqrt{Q}$ (slope -0.58, stable across ensembles) — while mapping two honest limits: the built-in failure alarm catches anti-correlated judges with no false positives yet can miss positively-correlated (shared-training) errors, and the two-solution tie-break inverts once judges are no longer clearly better than random — exactly where simple majority-voting evaluation, though biased, is the more robust fallback.

2 Introduction

Evaluation is the "forgotten twin" of learning: most AI work optimizes training, while the question *how good were these judges, really?* is usually answered only when labels exist. The unlabeled version of that question is old: Dawid-Skene-style observer-error models, annotator-expertise models, truth inference in crowdsourcing, and weak-supervision systems all use repeated judgments to infer reliability without a conventional answer key [Dawid and Skene, 1979, Yan et al., 2010, Zheng et al., 2017, Ratner et al., 2017]. Modern LLM-as-judge work makes the same problem newly practical and newly fragile: LLM judges can scale evaluation, but their errors, biases, and measurement uncertainty have to be modeled rather than wished away [Zheng et al., 2023, Chen et al., 2026].

That is also why majority vote and self-consistency are not enough as evaluation *instrumentation*. They can select an answer, but they do not by themselves tell us whether the voters were reliable, whether one judge was systematically wrong on one label, or whether the response matrix has enough variation to estimate reliability at all. A high-integrity evaluator needs to separate three questions that are often collapsed in practice: did the interface return valid measurements, is the vote matrix algebraically evaluable, and how does the selected estimate compare with external truth when such truth is available?

Algebraic evaluation (the **NTQR** logic [Corrada-Emmanuel et al., 2026, Corrada-Emmanuel, 2023a,b]; see the repository root README) takes a deliberately narrower stance. It constructs postulates — true for *any* finite test, containing no probability theory and no model of the task domain — that link the observed agreements and disagreements of an ensemble to the unknown statistics of their correctness. Agreement-rate estimators showed how much unlabeled signal is present in classifier agreements [Platanios et al., 2014, 2016]; NTQR uses a complete algebraic postulate system to ask when that signal is sufficient to recover judge accuracies, and when the response matrix itself warns that the evaluation is not justified.

A crucial caveat in that same framing is that the logic "is not magical … GIGO also applies to evaluation on unlabeled data," and its distinctive safety value is that it "can warn you if the ensemble is not good enough to perform a reliable evaluation." This paper turns that promise into a measured, reproducible demonstration on contemporary LLMs, and sharpens the warning into a quantitative diagnostic. The claim is not that NTQR solves crowdsourcing or judge aggregation generally. The narrow claim is that, under its stated algebraic assumptions, NTQR can recover per-judge accuracies from a finite unlabeled vote matrix — or emit an evaluability warning when the matrix lacks the required structure [Corrada-Emmanuel et al., 2024, Corrada-Emmanuel, 2025].

We ask three questions:

1. **Does unsupervised algebraic recovery actually work on real LLMs?** When an LLM ensemble *is* well-behaved, how close does the answer-key-free recovery come to the truth?
2. **What breaks it?** Across a spread of models — strong and deliberately weak — what failure mode dominates, and what is its mechanism?
3. **Can we predict failure without labels?** Is there a statistic, computable from the votes alone, that tells us in advance whether an ensemble is evaluable at all?

Our contributions: (i) an end-to-end, cached, reproducible pipeline that runs the same persona trio across many local models and scores unsupervised recovery against authored truth; (ii) the empirical finding that **disagreement is not evaluability** — the binding constraint is per-judge variation, not ensemble disagreement; and (iii) a label-free **evaluability diagnostic** that, in this study, perfectly predicted which models the algebra could solve.

The experiment is therefore not asking an LLM to predict markets better than another LLM. It is asking whether a finite matrix of binary judgments contains enough internal structure to evaluate the judges themselves. That distinction is what makes the method useful when a task model is unavailable: the algebra evaluates the response process, while authored truth is reserved for the controlled validation step.

2.1 Reader map: what is observed, recovered, and checked

Because the idea is unconventional, the central bookkeeping matters. The unsupervised evaluator never sees the authored truth; it sees only the vote matrix. The truth appears later, as a validation target for this controlled study. The simulations then ask whether the same algebra behaves the same way when the true data-generating process is known by construction.

Table 1: What is observed, recovered, and checked. The answer key is not an input to the unsupervised solve; it is the controlled-study check.

Object	Observed without labels?	Validation-only?	Meaning in this study
Scenario text	yes	no	A finite authored question containing only the market facts the model may use.
Persona vote	yes	no	One parsed `bullish` or `bearish` response, or an abstention if parsing fails.
Bullish / bearish labels	yes	no	Operational response labels: expected next-quarter direction up or down under the stated facts.
Authored truth	no	yes	The defensible answer used after the unsupervised solve to check recovery accuracy.
Prevalence	recovered	checked	Fraction of scenarios whose authored truth is `bullish` (`a`).
Persona accuracy	recovered	checked	Per-persona, per-label correctness rates inferred from vote-pattern structure.
Best predictor	recovered convention	checked	The higher-mean-accuracy algebraic branch selected without seeing truth.
Task/world model	no	no	Not estimated here; the algebra evaluates judges on a finite test rather than learning market semantics.

The important negative statement is as load-bearing as the positive one: without a task/world model, the algebra does **not** learn what financial markets will do. It evaluates a finite response matrix. In this study, `bullish` and `bearish` are operational labels defined by the authored scenario facts: `bullish` means the defensible next-quarter directional call is up under the stated facts; `bearish` means down. The estimator then asks whether the pattern of judge agreements is sufficient to recover each judge's per-label accuracy on that finite test.

Table 2: Representative operational questions and answers. The full 64-scenario answer key is generated in the supplemental; these rows show how `bullish` and `bearish` are defined without a learned task model.

ID	Operational truth	Why this is the authored answer	Scenario
s01	bullish	Growth, estimate beat, and raised guidance all point to upside.	Revenue grew 42% YoY, beat estimates, and management raised full-year guidance.

ID	Operational truth	Why this is the authored answer	Scenario
s02	bearish	Repeated misses plus a dividend cut indicate deteriorating fundamentals.	The company missed earnings for the third straight quarter and cut its dividend.
s19	bullish	Mixed but positive: the earnings beat is the operational tie-break despite full valuation.	Earnings beat but guidance was merely in line; shares are near all-time highs.
s20	bearish	Mixed but negative: margin compression and weak macro outweigh sales growth.	Sales grew but margins thinned and the macro outlook for the sector is murky.
s23	bullish	Mixed but positive: cost cuts plus reinvestment create a turnaround path.	New management is cutting costs aggressively and reinvesting in a growing niche.
s24	bearish	Mixed but negative: low valuation is outweighed by structural decline.	The stock is cheap on paper but sits in a structurally declining industry.

2.2 Related evaluation paradigms

The experiment is easiest to read if NTQR is placed beside the neighboring evaluation paradigms it resembles but does not replace. The table below keeps the comparison concrete: what is observed, what assumption does the work, what the method returns, and what kind of warning it can or cannot provide.

Table 3: Neighboring evaluation paradigms. Citations in the surrounding text identify the canonical references for each row; the key distinction is that NTQR returns either a label-free algebraic recovery or a label-free warning that recovery is not justified.

Paradigm	Observed input	Main assumption	Output	Failure warning
Supervised evaluation	Votes plus labels	Trusted labels	Empirical accuracies	None about unlabeled evaluability
Dawid-Skene / expertise	Repeated labels	Latent truth model	Worker error estimates	Model-fit warning
Agreement-rate estimation	Classifier agreements	Independence / priors	Accuracy estimates	Assumption sensitivity
Weak supervision	Labeling functions	Generative dependency model	Probabilistic labels	Dependency diagnostics
LLM-as-judge	Model judgments	Judge reliability model	Scalable evaluation	Bias / measurement-error correction
Majority vote / self-consistency	Multiple votes	Better-than-random voters	Aggregate answer	No per-judge recovery guarantee
NTQR in this paper	Unlabeled vote sketch	Algebraic postulates	Accuracies or alarm	Constant-classifier / imaginary-root failure

3 Methods

3.1 Evaluation object and data flow

The unit of analysis is a **model-specific vote matrix**: three personas by 64 scenarios, with each cell containing exactly one parsed `bullish` or `bearish` call. The unsupervised NTQR solve never consumes scenario text, market semantics, model logits, chain-of-thought, or the authored answer key. It receives only unlabeled vote-pattern counts derived from that matrix.

The controlled experiment adds two layers around that solve. First, a schema-constrained collection interface makes parsing a measured engineering boundary rather than a hidden source of invented labels. Ollama's structured output interface accepts a JSON Schema in the `format` field and its public guidance recommends lower-temperature decoding for more reliable schema adherence [Ollama, 2026b, 2024]. Structured-output reliability is itself an active evaluation problem [Wang et al., 2025]; here we make it a measured interface check and then move the scientific question to the completed binary vote matrix. Second, the authored scenario truth is kept out of the solve and used only afterward to score recovery error. fig. 1 shows this separation explicitly.

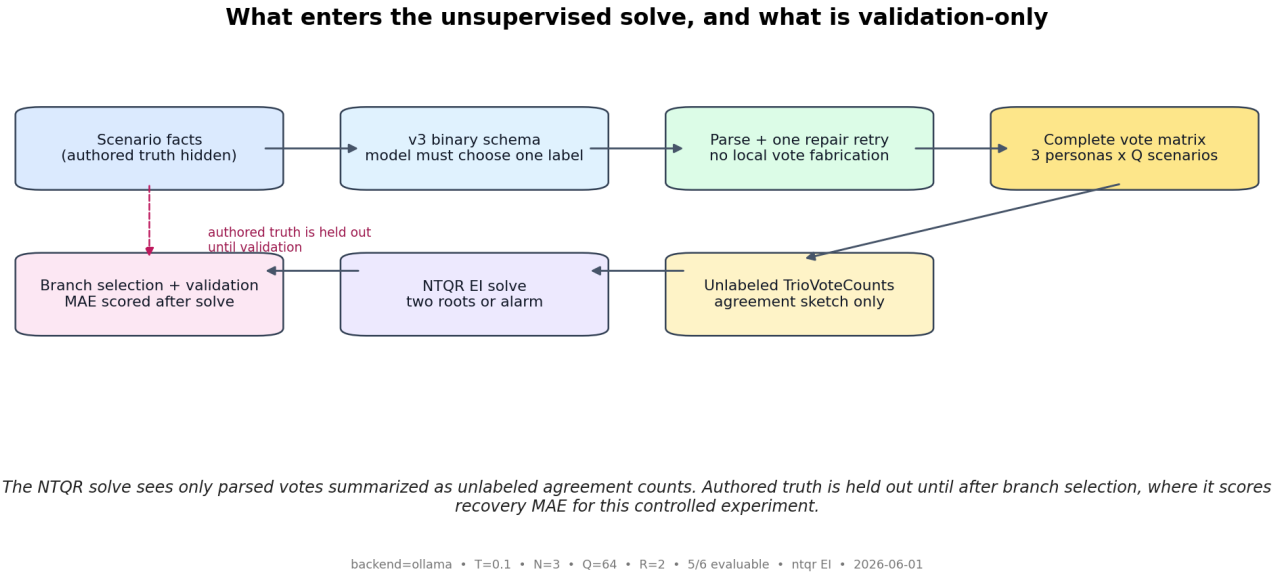


Figure 1: Pipeline boundary for the experiment. Scenario facts and system prompts create parsed persona votes; the unsupervised NTQR solve sees only the resulting unlabeled agreement sketch. Authored truth is held out until the validation step, where recovered accuracies are compared with supervised truth.

3.2 The classifiers: three LLM personas

We use three system-prompted "trader" personas as noisy binary judges, chosen so their *mistakes are not identical* — exactly the error structure error-independent evaluation needs:

- **Optimist** — leans `bullish` on ambiguous setups.
- **Neutral** — weighs the stated facts with no directional bias.
- **Pessimist** — leans `bearish` on ambiguous setups.

Three personas form a trio, matching the three-classifier `R=2` evaluators in the public NTQR package documentation [Corrada-Emmanuel et al., 2026]. Each persona reads the same scenario and returns a single JSON call. The prompt contract is versioned as `v3_binary_schema` (`4d157f84885a`) and appears verbatim in sec. 7.6. We treat the personas as instruction-following interfaces, not as human trader models; instruction-following behavior itself is a product of modern LLM alignment and fine-tuning practice [Ouyang et al., 2022].

3.3 The test: 64 binary scenarios

The test is $\mathbf{T = 1}$ test of $\mathbf{Q = 64}$ one-line market scenarios with $\mathbf{R = 2}$ possible responses (`bullish`→label `a`, `bearish`→label `b`). Each scenario carries an authored `truth` — the defensible call given only the stated facts. Most scenarios are clear-cut; six are deliberately ambiguous (mixed signals) so the personas' dispositions pull their calls apart.

The deck is deliberately *not* balanced. The authored prevalence is **40 bullish, 24 bearish** — true prevalence of label `a` = **0.625**. This is a methodological requirement, not an accident. The error-independent evaluator has a removable singularity at prevalence exactly 1/2: there its accuracy equations degenerate (the two algebraic solutions coalesce and the per-label accuracies resolve to 0/0, returned as `NaN`). An exactly 50/50 answer key places the *true* prevalence on that singularity, so any competent ensemble — whose recovered prevalence sits near the truth — lands on it and cannot be solved, even when every judge varies. Tilting the deck to 0.625 moves the whole problem off the singularity while leaving 24 clear bearish scenarios, so even the bullish-leaning personas keep casting both calls and never collapse into a constant classifier (sec. 4.2). The full scenario list and truths are in sec. 7.6.

The authored truth is **never** an input to the unsupervised evaluator. It is used only to build a `SupervisedEvaluation` as a check against which recovery is scored.

This distinction also defines the meaning of "best predictor" in the paper. The best model is not the model with the most plausible financial prose. It is the evaluable vote matrix whose selected unsupervised solution has the smallest validation-only recovery MAE against the authored labels.

3.4 The models

We run the identical trio through six locally-hosted chat models via Ollama at temperature 0.1 (stochastic decoding): `mistral:latest`, `gemma4:latest`, `gemma3:4b`, `gemma2:2b`, `granite4.1:3b`, `smollm2:135m-instruct-q4_K_S`. The models span a range of sizes and capabilities, from small sub-instruction-following models to multi-billion-parameter chat models. Votes are cached per `(backend, model, temperature, prompt version, prompt hash, persona, scenario)`, so the sweep is reproducible and resumable without mixing prompt designs.

The local Ollama tags are the experimental identifiers. Public model-family references provide provenance only: Mistral for `mistral:latest` [Jiang et al., 2023], the Gemma-family reports for the local Gemma tags [Gemma Team, 2024b,a, 2025], IBM Granite 4.1 for `granite4.1:3b` [IBM, 2026], and SmolLM2 as family context for the quantized local `smollm 2:135m-instruct-q4_K_S` tag [Ben Allal et al., 2025]. None of those citations enters the solve; the solve uses only the cached votes and the model names recorded in `summary.json`.

Table 4: Local model lineup, public family provenance, and data-driven run status. The citations identify model families, while the solve uses only cached votes and the recorded local tags.

Local Ollama tag	Public family citation	Run status	What the status means here
`granite4.1:3b`	IBM Granite 4.1 [IBM, 2026]	evaluable; recovery MAE 0.054	All three personas varied, every cell parsed, and the error-independent solve returned a selected branch.
`mistral:latest`	Mistral family [Jiang et al., 2023]	best recovered model; recovery MAE 0.012	All three personas varied, every cell parsed, and the error-independent solve returned a selected branch.
`gemma2:2b`	Gemma and Gemma 2 family [Gemma Team, 2024b,a]	evaluable; recovery MAE 0.026	All three personas varied, every cell parsed, and the error-independent solve returned a selected branch.
`gemma3:4b`	Gemma family reports [Gemma Team, 2024b, 2025]	evaluable; recovery MAE 0.024	All three personas varied, every cell parsed, and the error-independent solve returned a selected branch.
`gemma4:latest`	Gemma family reports [Gemma Team, 2024b, 2025]	evaluable; recovery MAE 0.024	All three personas varied, every cell parsed, and the error-independent solve returned a selected branch.
`smollm2:135m-instruct-q 4_K_S`	SmolLM2 family [Ben Allal et al., 2025]	not solved; constant persona(s): Optimist, Neutral, Pessimist	At least one judge never varied, so the trio carries no recoverable error signal for that judge.

Each requested model is asked the same binary task under the same schema. If a primary response violates the schema,

the collector allows one model-driven repair attempt using the same scenario facts and the same binary contract. A cell is accepted only if the model returns one of the two allowed labels; the collector never maps `neutral`, `mixed`, malformed JSON, or an unavailable model into a vote.

3.5 The evaluators

For each model we encode the trio's votes as `TrioLabelVoteCounts` and run three `ntqr.r2` evaluators [Corrada-Emmanuel et al., 2026, Corrada-Emmanuel, 2023b]:

- **`ErrorIndependentEvaluation`** (unsupervised) — recovers prevalence and each persona's per-label accuracy from agreement structure alone, with no answer key. It returns **two** algebraically consistent solutions (the labelling we want and its near-zero-accuracy mirror image).
- **`MajorityVotingEvaluation`** (unsupervised) — a simpler unsupervised baseline, also returning two solutions.
- **`SupervisedEvaluation`** (the check) — the answer-key truth.

Two-solution tie-break. From the two unsupervised solutions we keep the one with higher mean accuracy — the standard, answer-key-free "experts are better than random" assumption. We deliberately do *not* break the tie by closeness to the true prevalence: that would peek at the ground truth and would fail whenever the two prevalences are symmetric about 0.5. The derivation is in sec. 7.1.

3.6 The algebra: from votes to accuracies

This section makes the "exact algebra" concrete, because the whole study turns on its structure — including *why* it fails. The derivation follows the NTQR complete-postulate and streaming-algorithm line of work [Corrada-Emmanuel, 2023a,b], with the agreement-rate literature as its nearest statistical antecedent [Platanios et al., 2014, 2016]. We follow the construction in `SimplestExampleOfEvaluationWithAlgebraicGeometry.md` and the public NTQR implementation documented for `ntqr.r2.evaluators.ErrorIndependentEvaluation` [Corrada-Emmanuel et al., 2026].

The universal evaluation polynomial. Write the two labels as α (`bullish`, `a`) and β (`bearish`, `b`), the unknown test prevalence of α as P_α (with $P_\beta = 1 - P_\alpha$), and a single judge's per-label accuracies as $P_{i,\alpha}$ and $P_{i,\beta}$. The observed frequency with which that judge votes α decomposes — by definition, with no probability model — into "correct on α items" plus "wrong on β items":

$$f_\alpha \;=\; P_\alpha\, P_{i,\alpha} \;+\; P_\beta\,(1 - P_{i,\beta}). \tag{1}$$

This polynomial is *exact and universal*: any noisy judge on any finite test satisfies it identically. It contains no domain knowledge — that emptiness is precisely what makes it immune to the out-of-distribution failures of trained estimators. It is an evaluation *thermometer*, not a model.

The trio raises the rank. One judge gives two such equations in three unknowns — underdetermined. Three error-independent judges sharing one test supply enough agreement structure to close the system. The `ntqr` data sketch is the eight trio vote-pattern frequencies $f_{v_0 v_1 v_2}$ (each $v_i \in \{\alpha, \beta\}$, $\sum f = 1$). From them the evaluator forms three orders of statistic (shown here for label β; `datasketches.py`):

$$f_{\beta_i} = \sum_{v_j, v_k} f_{\ldots v_i = \beta \ldots}, \qquad \delta_{(i,j)} = f_{\beta_i \beta_j} - f_{\beta_i} f_{\beta_j}, \qquad m_{012} = \prod_i f_{\beta_i} + \sum_{(i,j,k)} f_{\beta_i}\, \delta_{(j,k)}. \tag{2}$$

Here f_{β_i} is judge i's marginal β-vote frequency, the **pair moment** $\delta_{(i,j)}$ is the (sample) covariance of judges i, j voting β — a *measured* agreement, not a modelled one — and m_{012} is the trio frequency moment. The prevalence of α is then the root of a **quadratic**

$$a\, P_\alpha^2 + b\, P_\alpha + c \;=\; 0, \qquad a = \Delta^2 + 4 \prod_{(i,j)} \delta_{(i,j)}, \quad b = -a, \quad c = \prod_{(i,j)} \delta_{(i,j)}. \tag{3}$$

where $\Delta = f_{\beta\beta\beta} - m_{012}$ is the gap between the all-β vote frequency and the trio frequency moment m_{012} (see `alpha_prevalence_quadratic_terms`). Because $b = -a$, the closed form collapses to a single $\pm$ pair:

$$P_\alpha \;=\; \tfrac{1}{2} \;\mp\; \tfrac{1}{2}\sqrt{1 - 4c/a}\,. \tag{4}$$

Two solutions, by construction. The $\mp$ is not numerical noise: absent labels, the votes alone admit *exactly two* logically consistent evaluation points — the labelling we want and its near-mirror image with the labels swapped. Each judge's per-label accuracy is then recovered as an exact *linear* function of P_α (`classifier_a_label_accuracy` / `classifier_b_label_accuracy`), so fixing the prevalence root fixes all six accuracy terms. The tie-break (sec. 3.5) selects between the two points without peeking at the truth.

The singularity at $P_\alpha = \frac{1}{2}$. The two roots coalesce exactly when $1 - 4c/a = 0$, i.e. when prevalence sits at $\frac{1}{2}$. There the radical vanishes, both solutions equal $\frac{1}{2}$, and the downstream accuracy ratios resolve to 0/0 (`NaN`). This is the removable singularity that motivates the deliberately unbalanced deck in sec. 3.3: an exactly 50/50 answer key puts the *true* prevalence on the singularity, so any competent ensemble lands on it and cannot be solved. Tilting prevalence to **0.625** moves the whole problem off the singular point.

The built-in alarm. When the judges are in fact error-*correlated* on the test, the discriminant $1 - 4c/a$ goes negative and the prevalence root becomes imaginary — `ntqr` surfaces this as a `PrevalenceImaginaryException`. An imaginary prevalence is an *iron-clad, label-free* detection that the error-independence assumption is violated: the algebra refuses to return a number it cannot justify. This is the "warning light" property at the center of the safety framing and the later no-knowledge alarm formulation [Corrada-Emmanuel et al., 2024, Corrada-Emmanuel, 2025] — and the per-judge constant-classifier failure of sec. 4.2 is its discrete cousin, which the following makes precise.

> **Proposition (per-judge evaluability).** *If any judge i is a constant classifier — it casts one label on all Q items — then the error-independent trio has no interior prevalence solution, regardless of the other two judges.*
>
> *Proof.* A constant judge has marginal $f_{\beta_i} \in \{0, 1\}$, so its sample variance is zero and **every pair moment touching i vanishes**, $\delta_{(i,j)} = f_{\beta_i\beta_j} - f_{\beta_i} f_{\beta_j} = 0$. Hence the product $\prod_{(i,j)} \delta_{(i,j)} = 0$, giving $c = 0$ and $a = \Delta^2$. The quadratic degenerates to $\Delta^2 P_\alpha(P_\alpha - 1) = 0$, whose only roots are the boundary points $P_\alpha \in \{0, 1\}$; the per-label accuracies then resolve to 0/0 and `ntqr` returns `NaN`. ■

This is why disagreement is the wrong axis. The degeneracy is triggered by a *per-judge* quantity ($\delta_{(i,\cdot)} = 0$ for a single i) and is blind to the *ensemble* quantity of how much the judges collectively disagree. Two judges can disagree violently while the third sits constant: the product still vanishes and the system is still unsolvable. Conversely the worked example in sec. 7.2 shows all three pair moments strictly positive on real votes, a positive discriminant, and the two mirror-image roots resolving to the truth. sec. 4.3 is the empirical face of this proposition; sec. 4.4 turns it into a label-free screen.

3.7 Statistics

- **Recovery MAE** (headline): mean absolute error between recovered and true per-persona, per-label accuracies. We also report it split by label.
- **Prevalence absolute error**: $|$ recovered $-$ true$|$ prevalence of label `a`.
- **Sampling-noise floor**: a reference scale for the recovery MAE. A per-label accuracy is estimated from only the questions carrying that label; the worst-case binomial standard error of such a proportion is `sqrt(0.25 / k)`, where `k` is the minority label's question count (here `k = 24`, giving a floor of **0.102**). A recovery MAE at or below this floor is smaller than the worst-case uncertainty of the supervised estimate it is compared against. We use it as a scale reference, not as a formal hypothesis test.
- **Degeneracy guard**: a constant classifier makes the algebra unsolvable and yields a SymPy `nan`; because `float(nan)` succeeds, we reject non-finite values explicitly (`math.isfinite`) and report the model as a *named failure* rather than averaging `NaN` into a meaningless statistic.
- **Scenario bootstrap**: uncertainty for the finite scenario deck is estimated by nonparametric resampling of scenarios with replacement, following the bootstrap logic of Efron and Tibshirani [Efron and Tibshirani, 1993].
- **Evaluability diagnostic** (label-free; see sec. 3.8).

3.8 The evaluability diagnostic

Computed from the votes alone — no answer key, no solve:

- **Vote variation** — for each judge, the fraction of scenarios on its single most common call (`modal_fraction`). A value of 1.0 means the judge never varied: a **constant classifier**, which carries no error signal.
- **Pairwise disagreement** — for each pair of judges, the fraction of scenarios on which they differ; and its mean over pairs.
- **Verdict** — `predicted_evaluable = (no judge is constant)`.

3.9 Reproducibility

Everything is exact and offline-reproducible from the cached votes. The numeric core (`scripts/llm/_analysis.py`) is pure — no I/O, no plotting — and covered by 99 unit tests that run with no network and no Ollama. The manuscript itself is *generated*: every number below is injected from `summary.json` by `scripts/manuscript/render.py`, so the prose cannot drift from the data. Exact commands are in sec. 7.7.

4 Results

4.1 Unsupervised recovery works when the ensemble is well-behaved

Among five evaluable models, mistral:latest recovered the authored truth most closely. Its unsupervised recovery tracked the authored truth to a mean absolute error of 0.012, within the 0.102 sampling-noise floor:

Table 5: Recovered vs. authored-truth per-persona, per-label accuracies for `mistral:latest`. The recovered columns use only votes; the truth columns are validation-only.

Persona	Label	True	Recovered	Abs. error
Optimist	a	1.0000	1.0000	0.0000
Optimist	b	0.7917	0.7600	0.0317
Neutral	a	0.9750	1.0000	0.0250
Neutral	b	1.0000	1.0000	0.0000
Pessimist	a	0.5750	0.5897	0.0147
Pessimist	b	1.0000	1.0000	0.0000

- **Recovery MAE = 0.012** (by label: `a` = 0.013, `b` = 0.011) across six accuracy terms.
- **Prevalence**: true 0.625, recovered 0.609; absolute error **0.016**.

The most striking row is the Pessimist on label `a`: the pessimist was genuinely *bad* at the bullish calls (true accuracy 0.57 — it called bearish even when the facts were clearly bullish), and the unsupervised algebra recovered that poor accuracy as 0.59 (against a true 0.57) — **with no answer key**. Unsupervised evaluation is not merely detecting who is good; it is quantifying who is bad. At temperature 0.1 the recovery is close but not exact (recovery MAE 0.012, within the 0.102 sampling-noise floor); §3.5 and the limitations frame this as one stochastic draw, not a general guarantee.

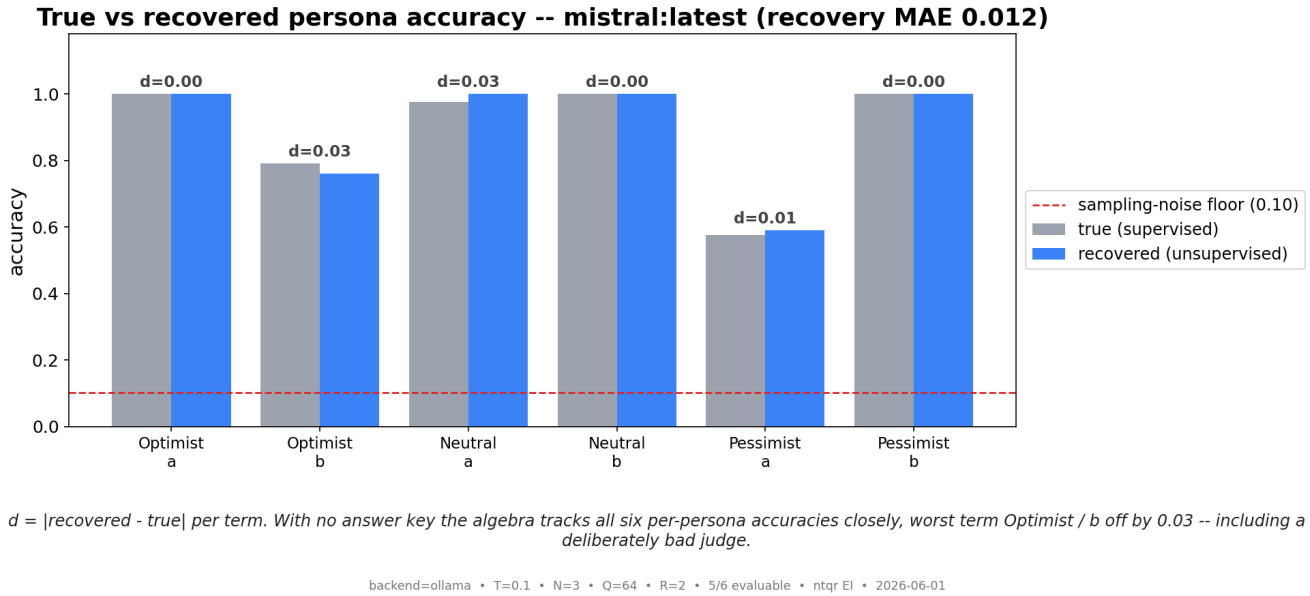
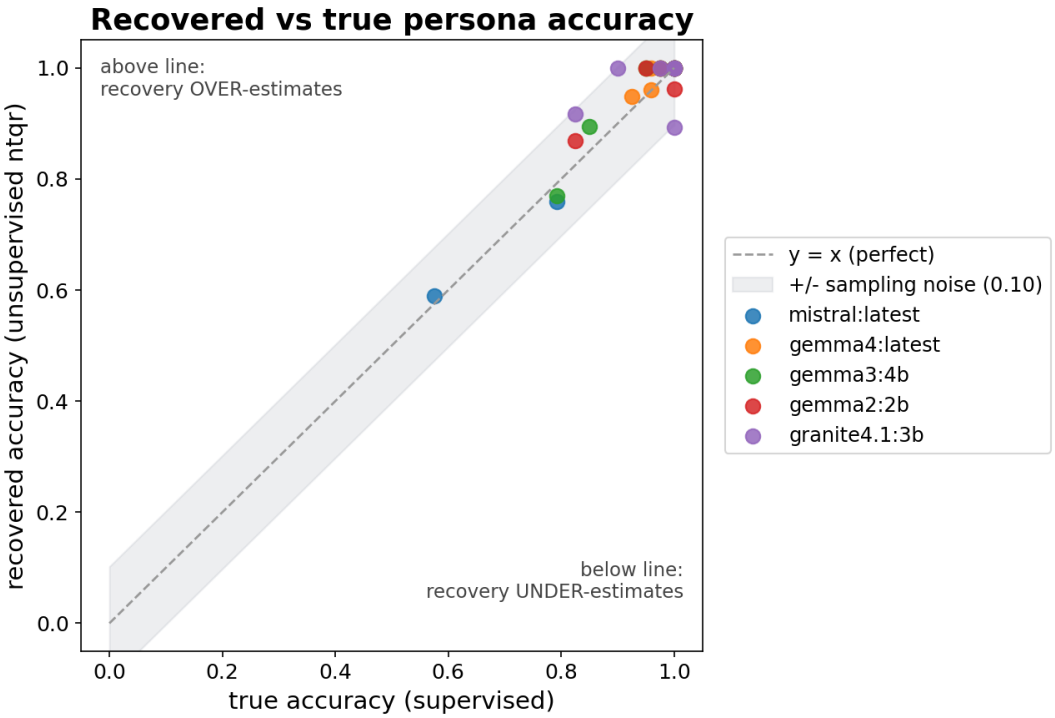


Figure 2: Recovered vs. authored-truth per-persona, per-label accuracy for `mistral:latest`. Bars track the truth term for term; the answer-key-free algebra recovers even the Pessimist's poor bullish accuracy (0.57). Recovery MAE = 0.012.

Across the five evaluable matrices, recovery MAE spans **0.012-0.054** with median **0.024** (tbl. 6). This range is useful context: NTQR is not merely ranking model quality, and a low recovery error does not mean every judge is good. It means the algebra recovered the judges' strengths and weaknesses faithfully. fig. 4 makes that distinction visible: the left panel shows validation-only true accuracies, while the right panel shows how much the unsupervised recovery missed each term by.



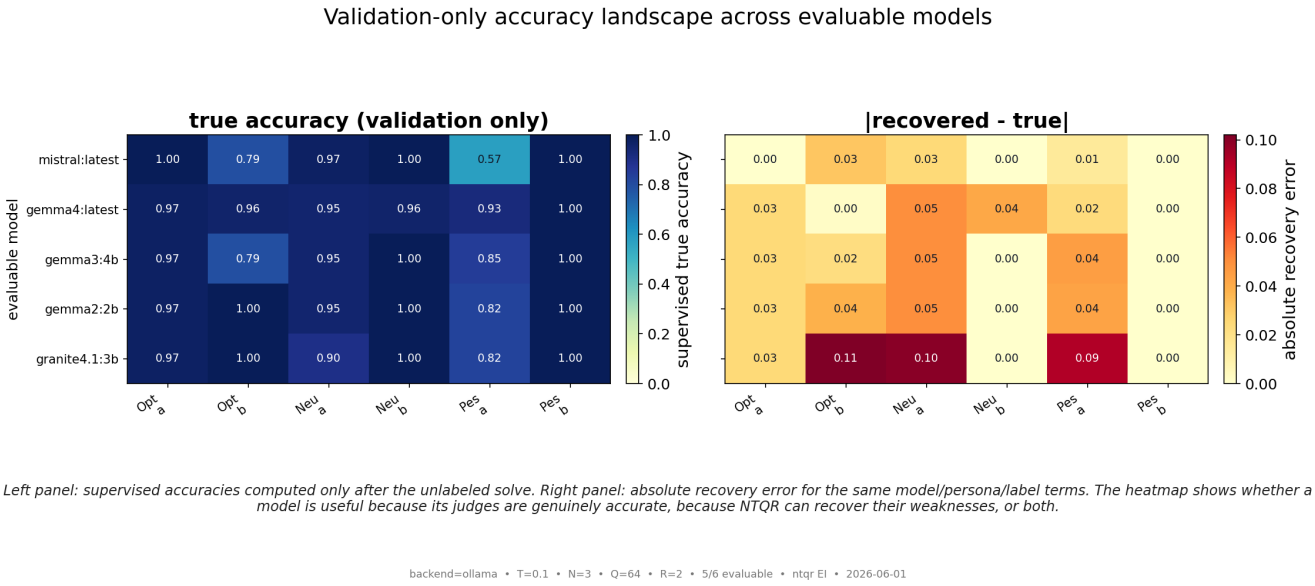
One point per persona/label accuracy term. Points on the dashed line are perfectly recovered; inside the grey band they are within one-question sampling noise of the supervised truth.

backend=ollama • T=0.1 • N=3 • Q=64 • R=2 • 5/6 evaluable • ntqr EI • 2026-06-01

Figure 3: The same recovery as a scatter of recovered vs. true accuracy: points on the diagonal have zero error. For the evaluable model the terms sit near the line (recovery MAE 0.012).

Table 6: Recovery ranking across evaluable model matrices. The MAE columns are computed after the unsupervised branch is selected; the authored truth is used only to score validation error.

Rank	Model	Recovery MAE	Max term error	Prevalence error	MV MAE	Validation role
1	`mistral:latest`	0.012	0.032	0.016	0.012	best recovered matrix
2	`gemma4:latest`	0.024	0.050	0.016	0.024	evaluable comparison
3	`gemma3:4b`	0.024	0.050	0.031	0.024	evaluable comparison
4	`gemma2:2b`	0.026	0.050	0.031	0.026	evaluable comparison
5	`granite4.1:3b`	0.054	0.107	0.062	0.054	evaluable comparison



Left panel: supervised accuracies computed only after the unlabeled solve. Right panel: absolute recovery error for the same model/persona/label terms. The heatmap shows whether a model is useful because its judges are genuinely accurate, because NTQR can recover their weaknesses, or both.

Figure 4: Validation-only accuracy landscape across evaluable models. The left panel shows supervised true accuracy by model/persona/label; the right panel shows absolute error of the unsupervised recovery for the same terms.

How stable is that number? The recovery MAE is one figure from one finite test, so we bootstrap it: resample the 64 scenarios with replacement and re-run the *same* unsupervised solve on each resample (2000 resamples, no new model calls). The point estimate is 0.012, with a 95% bootstrap CI of **[0.000, 0.038]** for the recovery MAE and [0.000, 0.047] for the prevalence error — the entire interval sits well below the 0.102 sampling-noise floor (fig. 5). In 2% of resamples a persona happened to collapse to a constant classifier and the solve was unavailable, a concrete measure of how the evaluability condition (not the recovery accuracy) is the fragile part.

4.2 Failure modes: constant classifiers and non-compliance

The other model collapsed at least one persona into a constant classifier:

Table 7: Per-model constant-classifier diagnostics. A model is evaluable only when every persona varies and every cell is parseable.

Model	Constant classifier(s)	Evaluable?
`granite4.1:3b`	—	**yes**
`mistral:latest`	—	**yes**

Model	Constant classifier(s)	Evaluable?
`gemma2:2b`	—	**yes**
`gemma3:4b`	—	**yes**
`gemma4:latest`	—	**yes**
`smollm2:135m-instruct-q4_K_S`	Optimist, Neutral, Pessimist	no

A capable model that follows a strongly-dispositioned persona prompt *too rigidly* turns that persona into a judge who answers the same way every time. A constant classifier carries no error signal, so the error-independent system becomes unsolvable (it returns a degenerate / `NaN` solution). We report these under `failed_models` with the offending persona named, rather than crashing the sweep or emitting a `NaN` statistic. fig. 6 shows each judge's modal-vote fraction per model; bars at 1.0 (red) are the constant classifiers — you can read the failures straight off the votes.

Non-compliance (no fabricated votes). At temperature 0.1, every model returned a parseable vote on all 192 persona calls – no responses were unparseable, so no model was dropped for non-compliance.

Table 8: Prompt-health diagnostics by model and persona under prompt version `v3_binary_schema`. Model labels are compacted for PDF fit; the full model names appear in tbl. 4 and in `summary.json`. Parse methods and entropy are derived from cached raw-response records.

Model	Persona	Primary	Repaired	Parsed	Invalid	Modal	Entropy	Parser
mistral	Optimist	1.000	0	64	0	0.703	0.877	schema:64
mistral	Neutral	1.000	0	64	0	0.609	0.965	schema:64
mistral	Pessimist	1.000	0	64	0	0.641	0.942	schema:64
gemma4	Optimist	1.000	0	64	0	0.625	0.954	schema:64
gemma4	Neutral	1.000	0	64	0	0.609	0.965	schema:64
gemma4	Pessimist	1.000	0	64	0	0.578	0.982	schema:64
gemma3	Optimist	1.000	0	64	0	0.688	0.896	schema:64
gemma3	Neutral	1.000	0	64	0	0.594	0.974	schema:64
gemma3	Pessimist	1.000	0	64	0	0.531	0.997	schema:64
gemma2	Optimist	1.000	0	64	0	0.609	0.965	schema:64
gemma2	Neutral	1.000	0	64	0	0.594	0.974	schema:64
gemma2	Pessimist	1.000	0	64	0	0.516	0.999	schema:64
granite	Optimist	1.000	0	64	0	0.609	0.965	schema:64
granite	Neutral	1.000	0	64	0	0.562	0.989	schema:64
granite	Pessimist	1.000	0	64	0	0.516	0.999	schema:64
smollm2	Optimist	1.000	0	64	0	1.000	-0.000	schema:64
smollm2	Neutral	1.000	0	64	0	1.000	-0.000	schema:64
smollm2	Pessimist	1.000	0	64	0	1.000	-0.000	schema:64

The previous temperature 0.7 prompt contract is retained as a stress-test baseline: it parsed 947/1152 cells, whereas the current temperature 0.1 schema-constrained headline parsed 1152/1152. The comparison isolates interface compliance from the algebraic evaluability question. fig. 8 shows why the low-temperature schema-constrained run replaces the temperature-0.7 prompt as the headline experiment: parseability is now an interface check, while evaluable vote matrices remain the scientific object.

4.3 Disagreement is *not* evaluability (the non-obvious result)

The intuitive story – "you need the judges to disagree" – is incomplete. In this low-temperature schema run, aggregate disagreement happens to separate the groups: the one unevaluable model sat at 0.00, while the five evaluable models spanned 0.03–0.23. That separation is a symptom of the failed model's constant-classifier collapse, not the algebraic gate.

Table 9: Mean pairwise disagreement by model. Disagreement is not the evaluability gate; per-judge variation is.

Model	Mean pairwise disagreement	Evaluable?
`smollm2:135m-instruct-q4_K_S`	0.00	no
`gemma4:latest`	**0.03**	**yes**

Model	Mean pairwise disagreement	Evaluable?
granite4.1:3b	**0.06**	**yes**
gemma2:2b	**0.06**	**yes**
gemma3:4b	**0.10**	**yes**
mistral:latest	**0.23**	**yes**

smollm2:135m-instruct-q4_K_S is unevaluable because one or more personas never varied, not because its ensemble-disagreement statistic fell below a universal threshold. The reason is structural: evaluability is gated by a **per-judge** necessary condition (every judge must vary), not by an **ensemble** quantity (how much the judges collectively disagree). A single judge who never varies breaks the algebra no matter how loudly the rest of the ensemble argues. Aggregate disagreement and per-judge variation are different axes, and only the latter is necessary for a solution to exist. fig. 9 plots the two axes against each other and shows the evaluable region is gated horizontally (by per-judge variation), not vertically (by disagreement).

4.4 A label-free diagnostic predicts evaluability

Because the binding condition is per-judge variation, it is detectable from the votes alone. Our evaluability diagnostic — flagging any judge whose modal-vote fraction is 1.0 — predicted `predicted_evaluable = True` for exactly gemma2:2b, gemma3:4b, gemma4:latest, granite4.1:3b, mistral:latest and `False` for the other one. The ntqr solve then evaluated exactly mistral:latest, gemma4:latest, gemma3:4b, gemma2:2b, granite4.1:3b. The prediction and the outcome agreed on all six models (`diagnostic_matched_outcome = true`). The diagnostic needs no answer key and no solve: it is computed directly from the response matrix, making it a usable pre-flight "evaluation thermometer."

4.5 Majority voting and error-independent recovery coincided here

For mistral:latest, `MajorityVotingEvaluation` and `ErrorIndependentEvaluation` returned the same selected solution, so their recovery MAE was identical (0.012). This is a property of this particular, near-error-independent vote pattern rather than a general theorem; we report both so the comparison is visible per run.

Majority voting is included because it is the classical aggregation baseline: Condorcet-style jury arguments justify it only when voters are independent and better than random [Dietrich and Spiekermann, 2021], and LLM self-consistency applies the same intuition by sampling multiple reasoning paths and taking the modal answer [Wang et al., 2022]. NTQR asks a different question: not which label wins, but whether the votes contain enough structure to estimate each judge's accuracy.

4.6 Reading the figure suite together

The 16 embedded figures tell the paper's arc in one pass. fig. 1 fixes the data boundary: votes enter the unsupervised solve, and authored truth enters only for validation. fig. 9 is the thesis: evaluability separates along the per-judge-variation axis, not the disagreement axis. fig. 2 and fig. 3 are the recovery result: for the best evaluable model, recovered and true accuracies track each other term for term (recovery MAE 0.012), while fig. 11 shows where every evaluable model sits. fig. 6 is the mechanism: the red 1.0 bars are exactly the judges that broke the algebra. The scorecard headline is: the five evaluable models sit at recovery MAE 0.012, within the 0.102 sampling-noise floor. Each figure ships a `<stem>.data.json` sidecar carrying its exact series and thresholds where applicable, so every plotted value is backed by a machine-readable number. fig. 4 separates two questions that should not be conflated: how accurate each judge actually was after validation, and how accurately the unlabeled solve recovered that fact. fig. 7, fig. 8, and fig. 10 expose the prompt interface and scenario difficulty that determine whether the algebra gets a usable response matrix; the remaining simulation figures extend the same logic through deterministic simulation, bootstrap uncertainty, the tie-break boundary, and the EI-vs-MV trade-off.

4.7 Synthetic validation: recovery scaling and the alarm's blind spot

The five evaluable real models give a finite empirical ranking, not a population estimate. To test the algebra across *many* ensembles — and to probe the failure detector directly — we ran a deterministic companion study on **synthetic judges** whose ground truth is set by construction (`scripts/sim/synthetic_recovery_study.py`; no network, fully seeded, reproducible). A trio with prevalence 0.55 and per-judge `(a, b)` accuracies (0.90, 0.80), (0.85, 0.75), (0.70, 0.95) is sampled under error-independence, its votes are fed through the **identical** `ErrorIndependentEvaluation` used on the LLM votes, and the unsupervised recovery is scored against the sample truth. We average 200 independent draws at each test size.

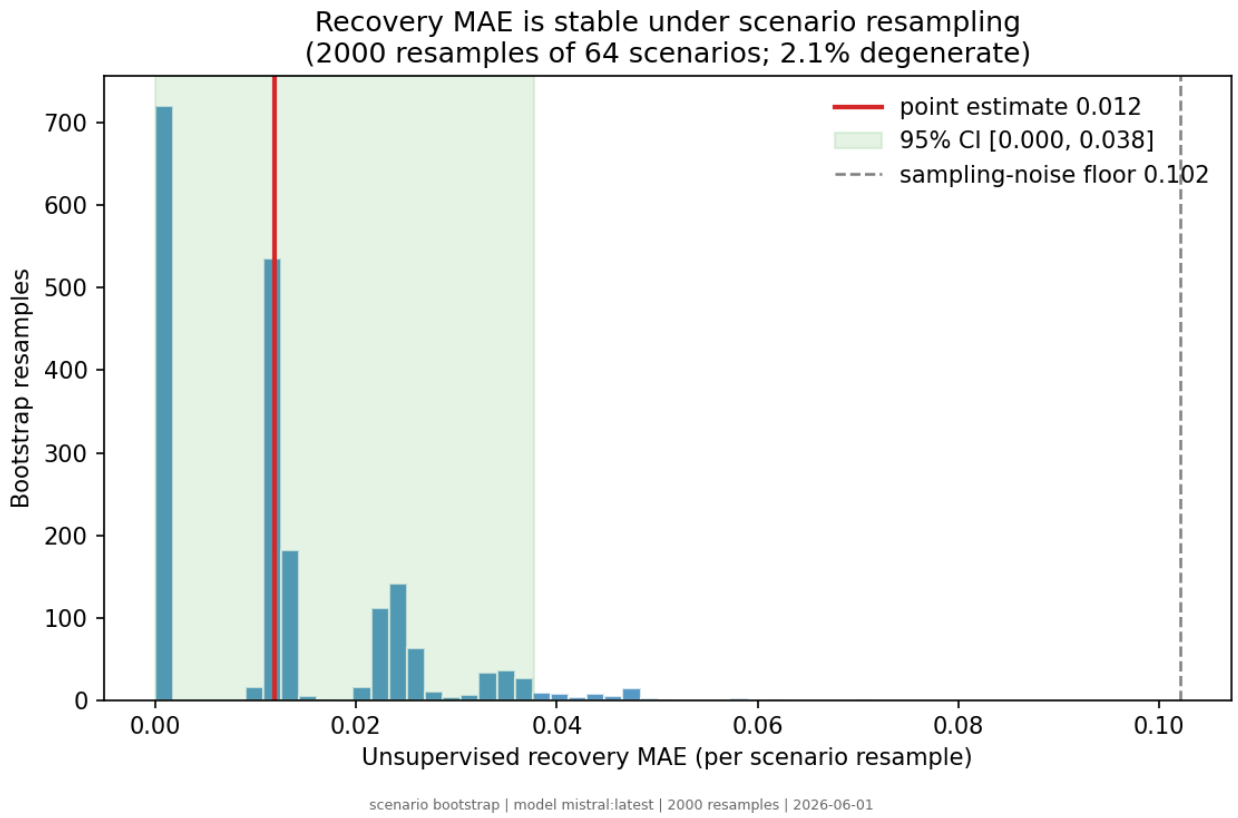


Figure 5: Bootstrap distribution of the unsupervised recovery MAE over 2000 scenario resamples of `mistral:latest`. The point estimate (0.012) and the 95% CI [0.000, 0.038] both sit far below the 0.102 sampling-noise floor; 2% of resamples were degenerate.

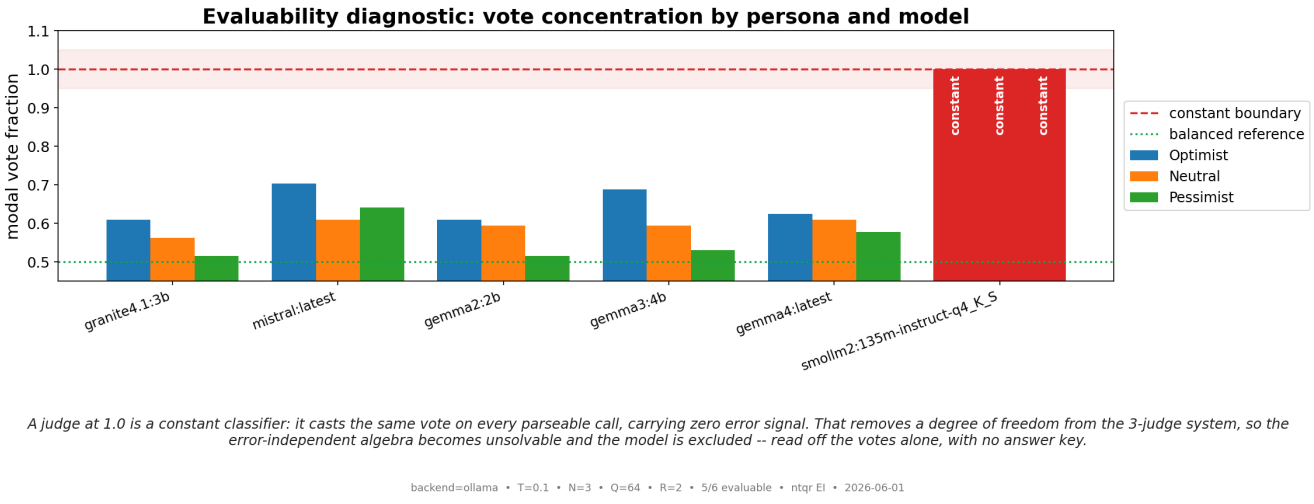
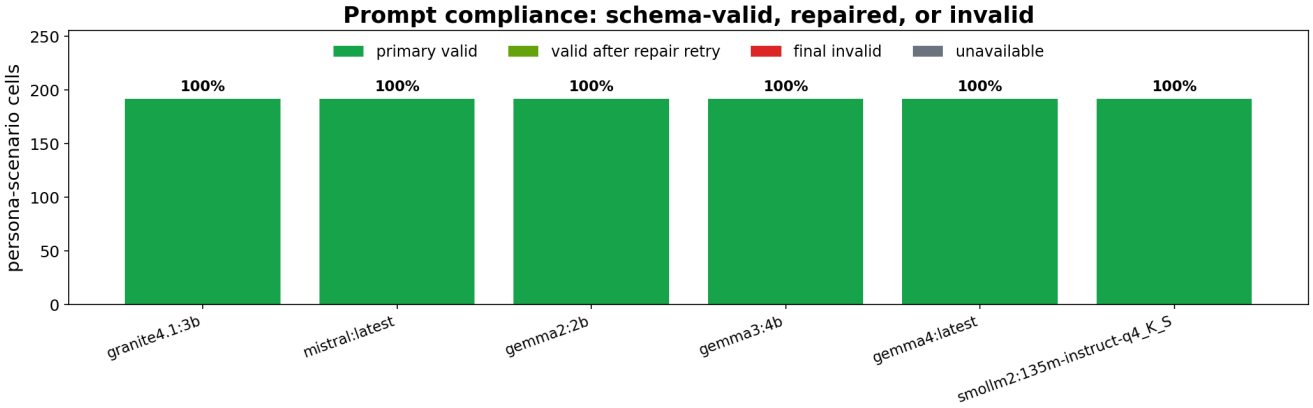


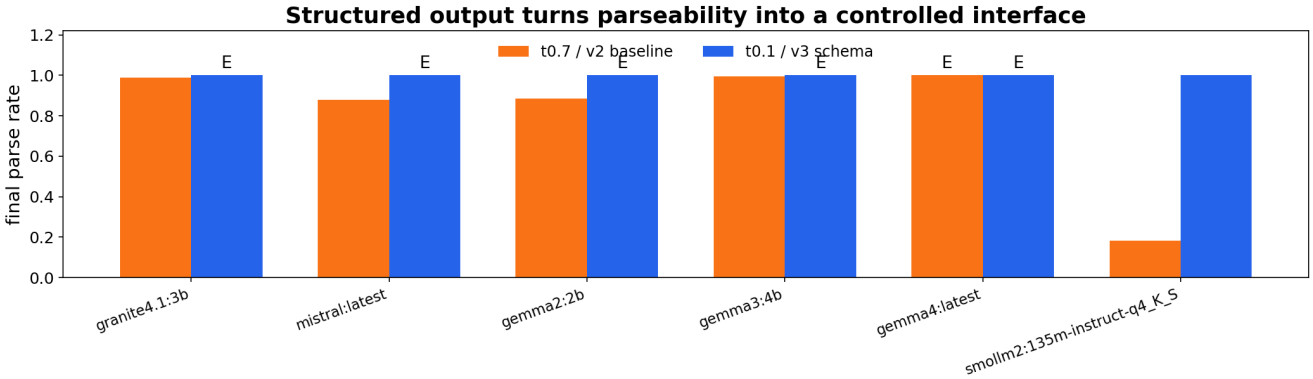
Figure 6: Per-judge modal-vote fraction for every model. A bar at 1.0 marks a constant classifier — a persona whose parseable calls never varied — which carries no error signal and makes the error-independent solve degenerate. The label-free read-out of which ensembles are unevaluable.



Each model is asked for the same persona-scenario grid under a binary JSON schema. Dark green cells were valid on the first attempt; light green cells became valid after one model-driven repair retry. Red and grey cells remain honest missing votes; no invalid response is converted into a fabricated vote.

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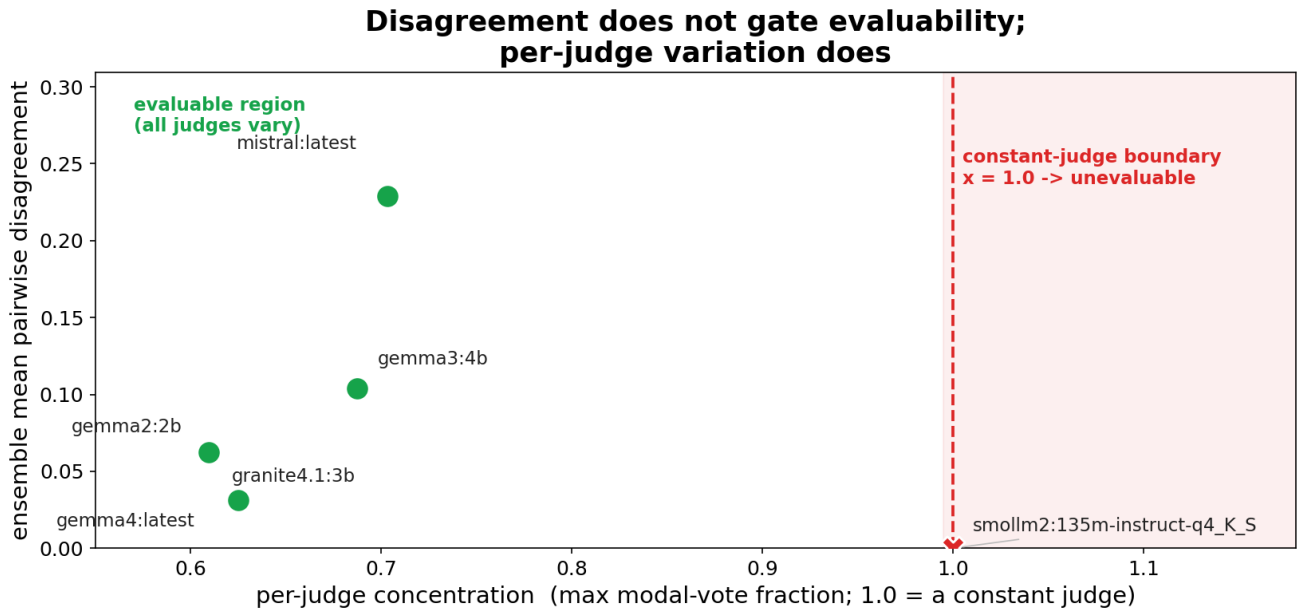
Figure 7: Prompt compliance for the persona-scenario grid. Dark green cells were valid on the first schema-constrained attempt; light green cells were accepted only after the one allowed model-driven repair retry; red/grey cells remain missing votes rather than fabricated labels.



The old temperature-0.7 prompt is retained as a stress-test baseline. The new temperature-0.1 schema run is the headline experiment: parseability should approach 100%, leaving genuine vote-matrix evaluability as the object of study. E marks an ensemble that the ntqr solve could evaluate.

backend=ollama • T=0.1 • N=3 • Q=64 • R=2 • 5/6 evaluable • ntqr EI • 2026-06-01

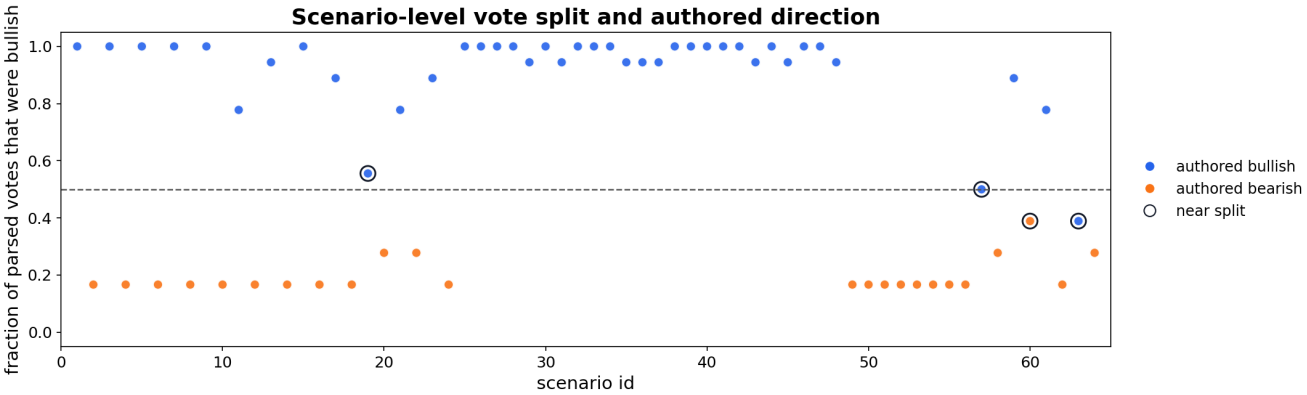
Figure 8: Temperature/schema comparison between the previous temperature-0.7 prompt contract and the current temperature-0.1 binary-schema run. Bar height is final parse rate; `E` marks a model whose final vote matrix was algebraically evaluable.



Each marker is one model. Green circles are evaluable; red crosses failed; 5/6 models are evaluable. The split is purely horizontal: any model at x = 1.0 has at least one constant judge, regardless of its ensemble disagreement (y).

backend=ollama • T=0.1 • N=3 • Q=64 • R=2 • 5/6 evaluable • ntqr EI • 2026-06-01

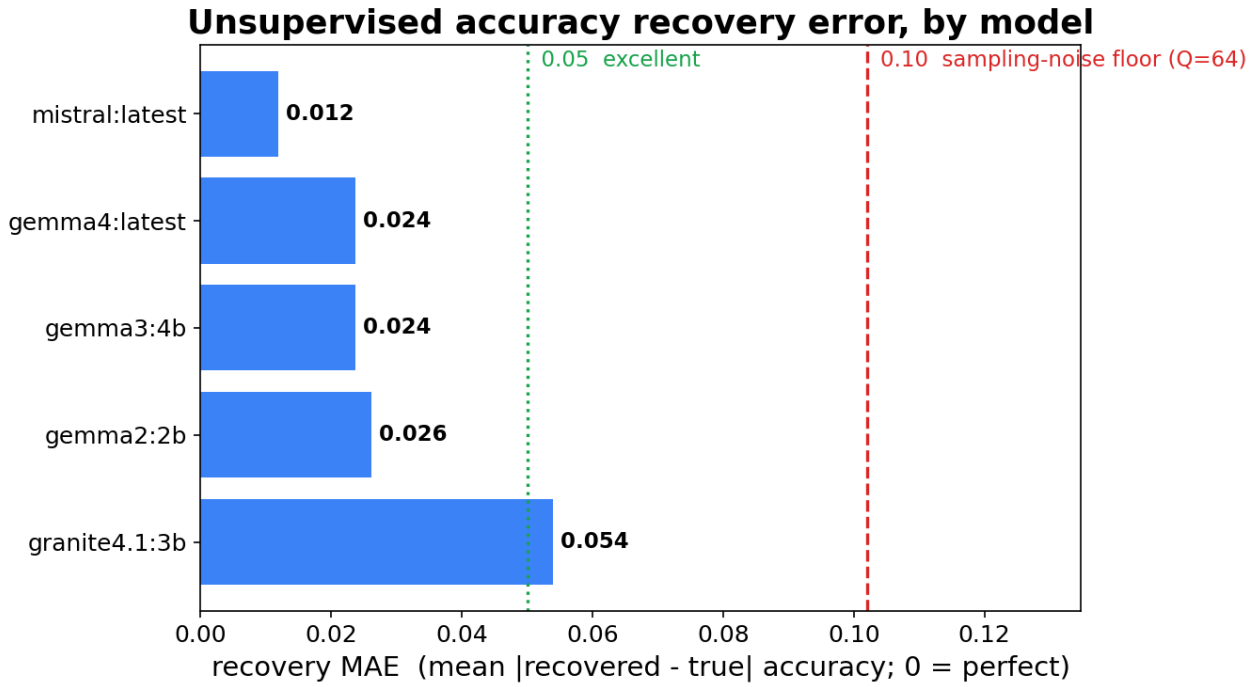
Figure 9: Per-judge modal-vote concentration (x, low = every judge varies) against ensemble mean pairwise disagreement (y). Evaluable models avoid the constant-classifier boundary; unevaluable models hit it. The y-axis is descriptive, while the x-axis contains the solvability condition.



Each point is one authored market scenario. The y-value is the fraction of parsed persona/model calls that were bullish; marker color is the authored operational truth. Points near 0.5 are the hard or intentionally mixed cases that create useful variation.

backend=ollama • T=0.1 • N=3 • Q=64 • R=2 • 5/6 evaluable • ntqr EI • 2026-06-01

Figure 10: Scenario-level vote split and authored direction. Each point is one authored market scenario. The y-value is the fraction of parsed persona/model calls that were bullish; marker color is the authored operational truth. Points near 0.5 are the hard or intentionally mixed cases that create useful variation.



Each bar is one model's mean |recovered - true| persona accuracy, recovered with no answer key. Left of the red line the error is within the conservative binomial sampling-noise reference for the minority label (24 of 64 questions).

backend=ollama • T=0.1 • N=3 • Q=64 • R=2 • 5/6 evaluable • ntqr EI • 2026-06-01

Figure 11: Scorecard: unsupervised recovery MAE for each evaluable model (lower is better), drawn against the "excellent" (0.05) and sampling-noise-floor (0.102) reference lines. the five evaluable models sit at recovery MAE 0.012, within the 0.102 sampling-noise floor.

Recovery error falls like sampling noise. As the test grows from `Q = 32` to `Q = 4096`, the mean recovery MAE drops from 0.123 to 0.007 (tbl. 10). A log–log fit of error against `Q` has slope **-0.58** ($R^2 = 0.989$; the prevalence error scales the same way, slope -0.55) — close to the −0.5 of an unbiased, sampling-noise-limited estimator. So the algebra carries no persistent bias: the only thing between the unsupervised recovery and the truth is finite-sample noise, and the real `mistral:latest` result (recovery MAE 0.012 at `Q = 64`) sits squarely on this curve rather than being a fortunate one-off.

Table 10: Synthetic recovery error vs. test size, averaged over 200 error-independent ensembles per row. Every draw solved cleanly — no spurious alarms.

Q	Accuracy MAE	Prevalence error	Trials solved
32	0.123	0.117	200/200
64	0.073	0.074	200/200
128	0.040	0.048	200/200
256	0.029	0.036	200/200
512	0.019	0.023	200/200
1024	0.013	0.016	200/200
2048	0.009	0.011	200/200
4096	0.007	0.008	200/200

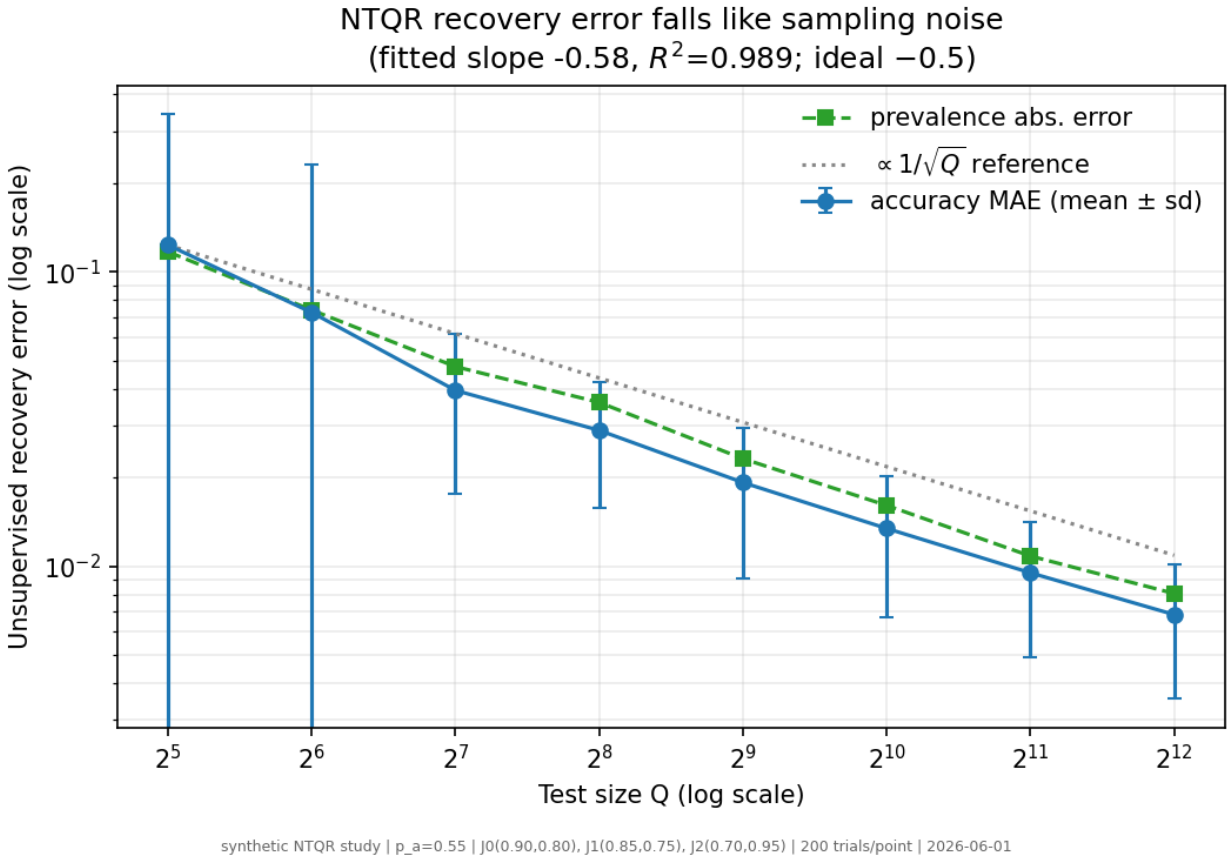


Figure 12: Unsupervised recovery error vs. test size `Q` on log–log axes, over 200 synthetic error-independent ensembles per point (accuracy MAE with ±1 sd error bars; prevalence error overlaid). The dotted line is a $1/\sqrt{\texttt{Q}}$ reference; the fitted slope -0.58 ($R^2 = 0.989$) matches the −0.5 of a sampling-noise-limited estimator.

The scaling is not an artifact of one ensemble. Repeating the convergence fit across 4 deliberately different ensembles — varied prevalence (0.50–0.70) and accuracy spread — keeps every fitted slope in a tight band **[-0.60, -0.50]** around the ideal −0.5 (tbl. 11), so the sampling-noise-limited behaviour is a property of the algebra, not of the particular judges we chose.

Table 11: The $1/\sqrt{Q}$ convergence slope is stable across diverse ensembles, each fitted over the same `Q` grid. All slopes cluster near the ideal −0.5.

Ensemble	Prevalence	Fitted slope	R^2
balanced p=0.50	0.50	-0.55	0.999
tilted p=0.70	0.70	-0.51	0.995
strong p=0.55	0.55	-0.50	0.997
modest p=0.65	0.65	-0.60	0.986

The alarm is sound but has a blind spot. The error-independent solve carries a built-in failure detector: when the prevalence quadratic's discriminant `D = 1 − 4c/a` goes negative the prevalence root is imaginary — an answer-key-free signal that error-independence is violated. We characterised it by injecting two kinds of dependence at `Q = 1024` (tbl. 12, fig. 13). On genuinely error-independent draws the alarm fired in **0%** of trials — no false positives. An **anti-correlated** judge pair (which drives a pairwise vote-moment negative) tripped it in up to **94%** of trials. But positive **common-mode** correlation — all judges sharing a forced error, the natural analogue of shared training data — never tripped it (**0%**), even as it silently worsened the recovery it failed to warn about (prevalence error rising from 0.016 under independence to 0.042 at the strongest common-mode setting).

Table 12: Alarm-firing rate by error structure at `Q = 1024` (200 trials each). `D` is the mean prevalence discriminant; the last column is the recovery error on the trials where the alarm stayed silent.

Error structure	Alarm fires	Mean D	Recovery error (alarm silent)
independent	0%	0.0115	0.016
common-mode 0.1	0%	0.0080	0.014
common-mode 0.2	0%	0.0041	0.024
common-mode 0.3	0%	0.0021	0.034
common-mode 0.4	0%	0.0013	0.042
anti-pair 0.2	0%	0.0126	0.043
anti-pair 0.3	40%	0.0962	0.204
anti-pair 0.4	94%	-0.0726	0.407

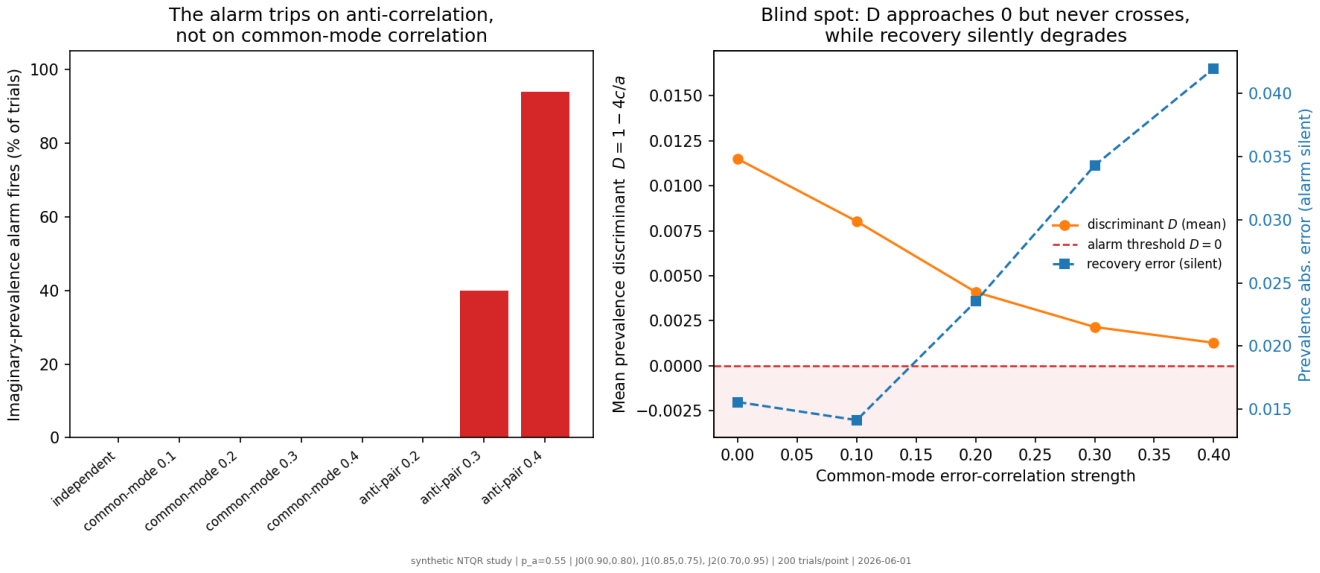


Figure 13: Left: the imaginary-prevalence alarm fires only under anti-correlation, never on independent or common-mode data. Right: the mechanism — as common-mode correlation strengthens, the discriminant `D` falls toward the `D = 0` alarm threshold but never crosses it, while the un-warned recovery error climbs.

This sharpens the safety claim honestly: the imaginary-prevalence alarm is a *sufficient*, never-false-positive detector of violated error-independence, but it is **not necessary** — the most realistic LLM failure, positively-correlated errors from

shared pre-training, is exactly the regime it can miss. The per-judge evaluability diagnostic (sec. 3.8) and this alarm are complementary screens, and neither is a substitute for a held-out check when correlated errors are plausible.

4.8 The two-solution tie-break needs clearly-better-than-random judges

Error-independent evaluation returns two mirror-image solutions; we pick the one with higher mean accuracy (sec. 7.1), the standard better-than-random assumption. That assumption has teeth, and a synthetic sweep shows exactly where it loses them. We hold the trio symmetric (so mean per-label accuracy is the only varying factor), fix prevalence 0.60 and `Q = 256`, and slide the accuracy from strong toward random (tbl. 13, fig. 14). For strong judges the tie-break is essentially flawless — **0%** catastrophic mirror-selection down to accuracy **0.82** — but as the judges approach random the mirror starts winning by chance and recovery inverts, reaching **85%** catastrophic selections near accuracy 0.5. The practical rule: algebraic recovery is trustworthy only for an ensemble that is *clearly* better than random; a barely-better-than-chance jury can be confidently and silently inverted.

Table 13: Catastrophic mirror-selection (prevalence error > 0.25) vs. mean judge accuracy at prevalence 0.60, `Q = 256`, 200 trials each.

Judge accuracy	Catastrophic mirror-selection	Mean prevalence error
0.90	0%	0.014
0.82	0%	0.039
0.76	4%	0.093
0.70	26%	0.185
0.62	71%	0.421
0.56	85%	0.485
0.52	82%	0.487

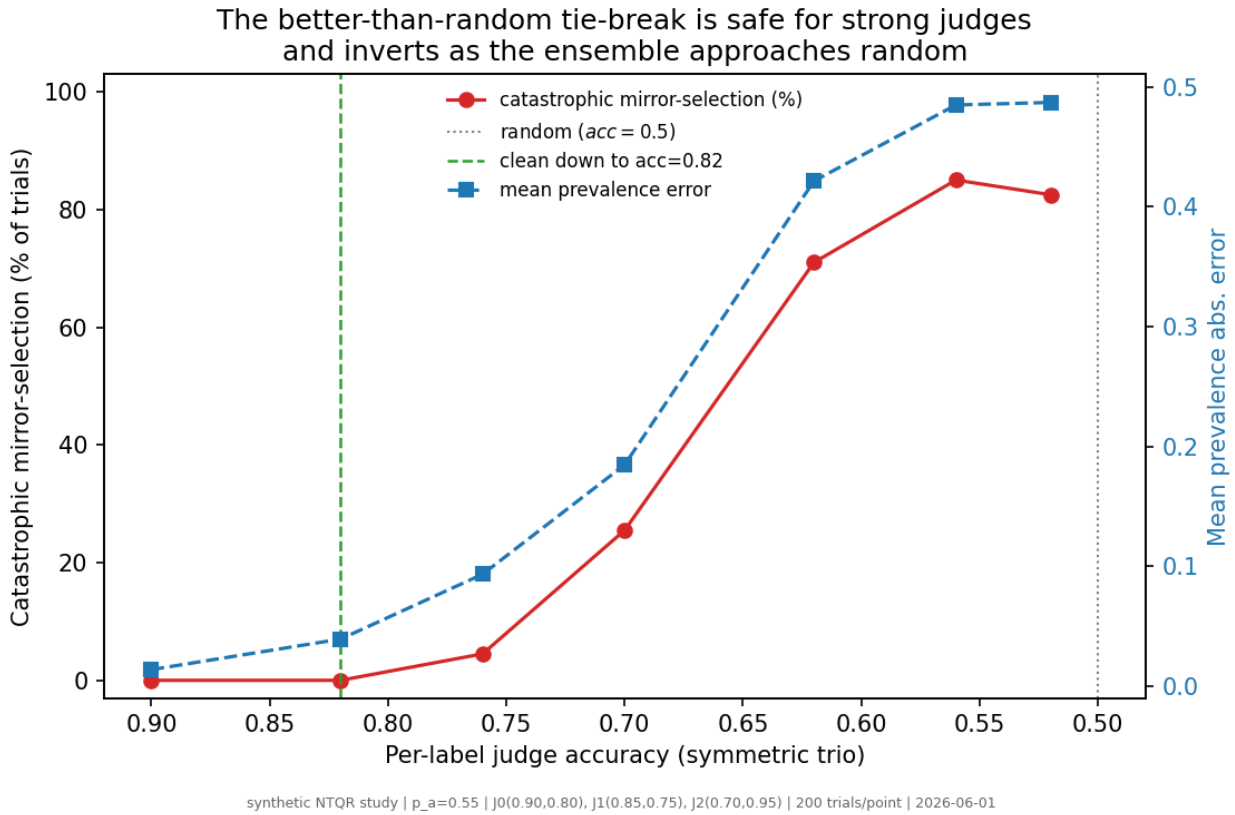


Figure 14: The better-than-random tie-break is flawless for strong judges and inverts as the ensemble approaches random: catastrophic mirror-selection climbs from 0% to 85% and the mean prevalence error rises with it.

4.9 Error-independent vs. majority-voting: exactness vs. robustness

sec. 4.5 reported that on `mistral:latest` the error-independent (EI) and majority-voting (MV) evaluators returned the same recovery — explicitly *"a property of this particular vote pattern, not a general theorem."* The synthetic harness settles the general question by running **both** evaluators on the same draws across a prevalence × accuracy grid (`Q = 512`; tbl. 14, fig. 15). The answer is a clean trade-off, not a winner:

- **EI is more accurate where its assumption holds.** For clearly-better-than- random judges the exact algebra beats majority voting — e.g. at prevalence 0.55, accuracy 0.74, EI's recovery MAE is 0.047 vs MV's 0.068 (1.43× lower), and EI wins on 79% of draws. MV is *biased* — it assumes the majority vote is the truth — so it cannot reach the exact answer.
- **MV is more robust where EI's assumption fails.** In the tie-break danger zone of sec. 4.8 (weak judges, here accuracy 0.58 at prevalence 0.70), EI's mirror inverts and its MAE explodes to 1.098 while MV — which has no imaginary roots and no mirror — stays at 0.176; EI wins only 18% of draws.

The practical guidance mirrors the diagnostics of sec. 3.8 and sec. 4.8: prefer the exact algebra for an ensemble that clears the evaluability bar and is clearly better than random, but keep majority voting as the robust fallback when those conditions are in doubt — and report both, as we do per run.

Table 14: Error-independent vs. majority-voting recovery MAE across a prevalence × accuracy grid (`Q = 512`, 200 trials per cell). "EI wins" is the fraction of draws where the exact algebra's MAE was lower.

Prevalence	Judge accuracy	EI MAE	MV MAE	EI wins
0.55	0.90	0.011	0.011	60%
0.55	0.82	0.023	0.032	83%
0.55	0.74	0.047	0.068	79%
0.55	0.66	0.841	0.116	61%
0.55	0.58	1.235	0.176	25%
0.70	0.90	0.012	0.013	65%
0.70	0.82	0.027	0.031	66%
0.70	0.74	0.059	0.065	64%
0.70	0.66	0.357	0.113	47%
0.70	0.58	1.098	0.176	18%
0.85	0.90	0.018	0.019	61%
0.85	0.82	0.041	0.037	50%
0.85	0.74	0.096	0.063	45%
0.85	0.66	1.589	0.109	25%
0.85	0.58	1.432	0.173	20%

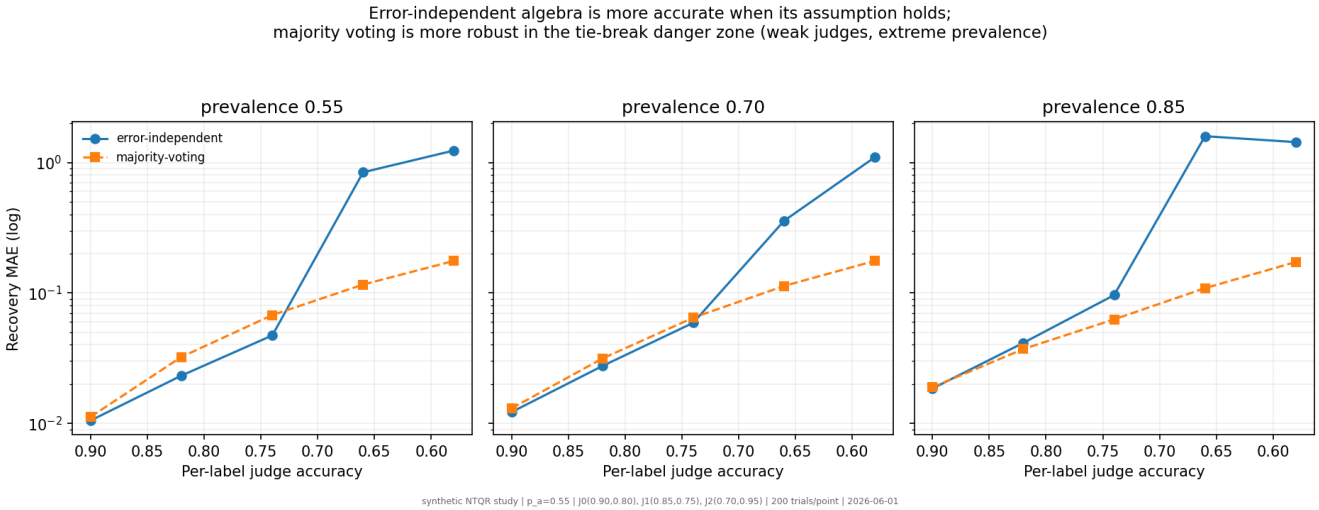


Figure 15: Recovery MAE of the error-independent algebra vs. majority-voting as judge accuracy falls from strong to near-random, one panel per prevalence. EI is lower while judges are strong, then crosses above and diverges as its tie-break inverts; MV stays bounded throughout.

4.10 Statistical validation matrix

The manuscript's main claims are checked by a generated validation report rather than by prose alone. `scripts/validation/statistical_validation_report.py` reads the LLM summary, simulation summaries, bootstrap summary, and figure sidecars; it does not call models or rerun experiments. The report currently passes **9/9** checks, covering the real recovery scale, label-free diagnostic, bootstrap uncertainty, synthetic convergence, alarm behavior, tie-break boundary, EI-vs-MV trade-off, and figure sidecar completeness.

Table 15: Offline validation checks for the manuscript's load-bearing claims. Each row is computed from generated JSON artifacts, not hand-audited numbers in prose.

Check	Result	Evidence
Label-free diagnostic matches solver outcome	PASS	predicted 5; evaluated 5
Real recovery is below sampling-noise floor	PASS	MAE 0.028 <= floor 0.102
Bootstrap CI remains below sampling-noise floor	PASS	CI_hi 0.038 <= floor 0.102
Synthetic recovery follows 1/sqrt(Q)	PASS	slope -0.585; R2 0.989
Imaginary-root alarm has zero independent false alarms	PASS	false alarm 0%
Common-mode dependence is reported as a blind spot	PASS	alarm 0%; silent error 0.042
Better-than-random tie-break boundary is quantified	PASS	safe to 0.82; max inversion 85%
EI-vs-MV trade-off has both regimes	PASS	EI win 79%; danger-zone EI win 18%
Every source data figure has a parseable sidecar	PASS	16/16 sidecars parse

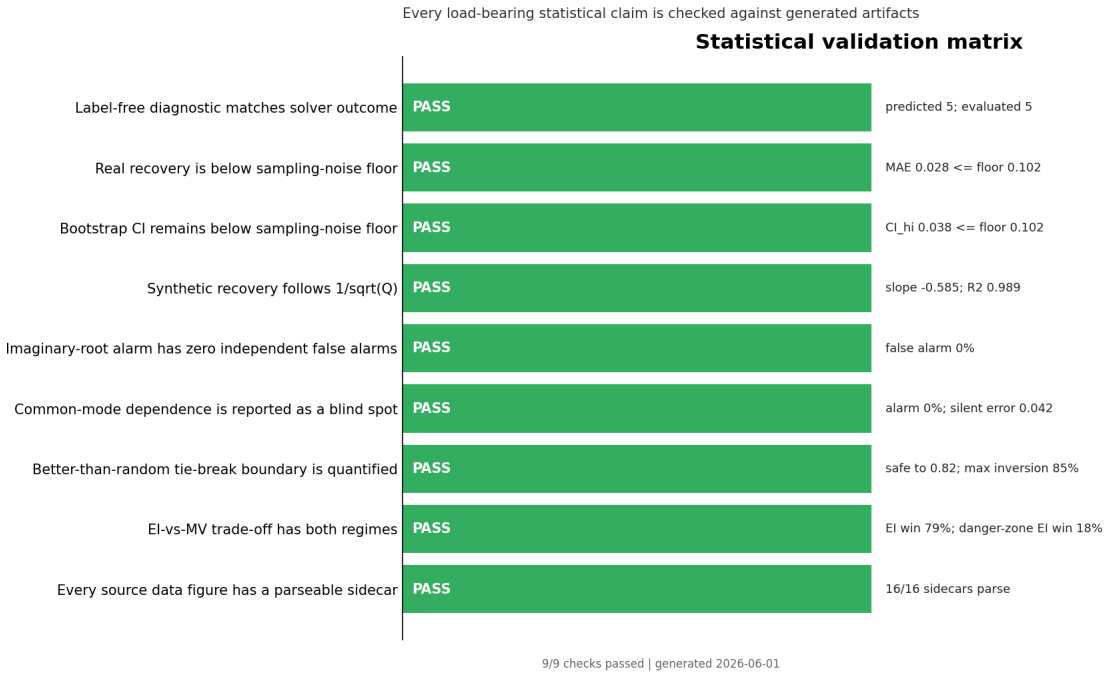


Figure 16: Validation matrix for the statistical claims. Green rows are checks that passed against the generated artifacts; the evidence labels show the actual thresholds or counts used by each check.

5 Discussion

What this shows. On real, contemporary LLM-persona ensembles that satisfy the evaluability condition, purely algebraic, answer-key-free evaluation recovered per-judge accuracies closely (recovery MAE 0.012, within the 0.102 sampling-noise floor) – including correctly identifying and quantifying a *bad* judge. That is the positive existence result: the NTQR logic is not only theoretically universal but operationally effective on real model vote matrices, while still refusing to solve a matrix whose judges carry no variation signal.

How it fits the literature. The result is complementary to crowdsourcing, weak-supervision, and LLM-as-judge measurement-error methods rather than a replacement for them [Zheng et al., 2017, Ratner et al., 2017, Zheng et al., 2023, Chen et al., 2026]. Those approaches ask how to infer or aggregate task truth under a model of workers, labeling functions, or judges; majority-vote and self-consistency baselines ask which answer should win under better-than-random aggregation assumptions [Dietrich and Spiekermann, 2021, Wang et al., 2022]. This experiment asks a prior algebraic question: before using a judge ensemble as an evaluator, does the unlabeled response matrix contain the variation and agreement structure required for evaluation at all? That is the practical role of the NTQR alarm lineage [Corrada-Emmanuel et al., 2024, Corrada-Emmanuel, 2025].

The sharper lesson is the warning. The repository frames the safety value of algebraic evaluation as its ability to "warn you if the ensemble is not good enough." We make that warning concrete and show it fires correctly: one of six models was not evaluable, and a label-free diagnostic predicted it in advance. Crucially, the warning is *not* "did the ensemble disagree enough?"; the failing model sits at low disagreement because its personas collapsed into constant classifiers. The correct warning is per-judge: *at least one of your judges is a constant classifier.* For practitioners assembling LLM juries, this reframes ensemble design: maximizing diversity/disagreement is neither sufficient nor the right target; ensuring every judge actually exercises both responses across the test is the necessary condition.

Implications for evaluation engineering. The operational lesson is to treat LLM evaluation as measurement engineering, not just answer aggregation. First, make the interface boring: schema-constrained, low-temperature collection and explicit repair diagnostics should make parseability a controlled boundary, not the headline result. Second, check the unlabeled matrix before trusting any recovered score: a constant judge is not a weak signal, it is no signal for the trio algebra. Third, distinguish *good judges* from *recoverable judges*: fig. 4 shows that the scientific value is not only finding high accuracies, but recovering poor or label-specific accuracies without looking at the answer key. That is what makes the method relevant to LLM-as-judge systems where the evaluator itself is part of the risk surface.

Why strong models fail this way. A strongly dispositioned persona prompt is a recipe for a constant classifier: a model follows its instruction ("lean bullish") so rigidly that it casts the same call on every scenario. The very capability that makes a model a good instruction-follower can make it a useless *evaluable* judge — in this sweep a capable mid-sized model collapsed personas into constant classifiers (sec. 4.2). Crucially this is *not* cured by sampling: it appears here even at temperature 0.1 (stochastic decoding), so the failure is prompt-induced degeneracy, not a decoding artifact. This is a caution about over-prompting LLM-as-judge ensembles.

Limitations (and what would falsify the claims).

- **Single run, single test.** `T = 1`, `Q = 64` — a single stochastic draw at temperature 0.1. The recovery result is one sample, not a population estimate; because decoding is stochastic, repeating the sweep would now yield a distribution of recovery MAE (a fixed temperature-0 run could not). A stronger design would repeat the draw many times and report that distribution, and also vary the scenario set. Decoding is fixed, but the scenario set is itself a finite sample, so sec. 4.1 bootstraps over it: the recovery MAE carries a 95% CI of [0.000, 0.038] and 2% of resamples are degenerate — the test-draw uncertainty this limitation flags, now quantified.
- **Recovery needs clearly-better-than-random judges.** The two-solution tie-break (sec. 7.1) assumes the ensemble beats random; sec. 4.8 shows it stays exact down to accuracy 0.82 but inverts (up to 85% catastrophic) as judges approach chance. The evaluated `mistral:latest` judges are well clear of that boundary, but a borderline jury is not safe.
- **Five evaluable models.** `mistral:latest`, `gemma4:latest`, `gemma3:4b`, `gemma2:2b`, `granite4.1:3b` cleared the evaluability bar, so the head-to-head recovery ranking compares five real models. The recovery-quality claim rests on those finite real vote matrices; the *failure-mode* and *diagnostic* claims rest on all six. The deterministic synthetic study (sec. 4.7) extends the recovery-quality claim across 200 ensembles per test size where the ground truth is known by construction, showing the real result is on-curve, not a fluke.
- **Diagnostic is necessary, not proven sufficient.** "No constant classifier" is a *necessary* condition for the error-independent solve; this study shows it was also sufficient on these six models, but six models is not a powered test of sufficiency. A model could in principle vary on every judge yet still violate error-independence and recover poorly. The honest claim is: the diagnostic is a cheap, label-free *necessary-condition* screen that caught every failure here.

- **Authored truth.** "Truth" is the authors' defensible call, not a market outcome; it calibrates the check, not the unsupervised recovery.

Future work. Repeat across temperatures and scenario draws to get a recovery distribution; extend beyond the trio (larger `N`, `R>2`); test whether a softened persona prompt (or light temperature) restores variation in the collapsed judges and thereby restores evaluability. The synthetic study (sec. 4.7) begins the last of these — it shows positive common-mode correlation silently degrades recovery without tripping the alarm — and a natural next step is to *measure* the realized error-correlation of real LLM juries (e.g. from shared base models) and predict the resulting recovery bias.

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6 Figures — formal specification

Every figure is regenerated by the pipeline — the real-sweep figures into `outputs/llm/multi_llm_personas_evaluation/`, the simulation figures into `outputs/sim/synthetic_recovery_study/`, the bootstrap figure into `outputs/sim/bootstrap_recovery_ci/`, and the validation figure into `outputs/validation/statistical_validation/` — and ships a machine-readable `<stem>.data.json` sidecar holding its exact series, threshold values, and a provenance block (backend, model set, temperature 0.1, Q=64, render date). A figure is therefore not an illustration but a *typeset view of a data file*: the claim, the encoding, and the numbers are all pinned. Each figure below is specified as **claim -> encoding -> reference lines -> data source**.

- `fig:disagreement-vs-evaluability` (`disagreement_vs_evaluability.png`; `disagreement_vs_evaluability.data.json`) supports sec. 4.3 and sec. 3.6. It plots per-judge modal-vote concentration against mean pairwise disagreement, with a vertical evaluability band.
- `fig:evaluation-pipeline` (`evaluation_pipeline.png`; `evaluation_pipeline.data.json`) supports sec. 3.1. It shows which objects are observed, which are inferred by NTQR, and where authored truth enters only as a validation target.
- `fig:best-model-recovery` (`best_model_recovery.png`; `best_model_recovery.data.json`) supports sec. 4.1. It uses grouped bars for recovered vs. true per-(persona, label) accuracy, with the sampling-noise floor.
- `fig:recovered-vs-true` (`recovered_vs_true_scatter.png`; `recovered_vs_true_scatter.data.json`) supports sec. 4.1. It plots true accuracy against recovered accuracy with the $y = x$ identity line.
- `fig:validation-accuracy-heatmap` (`validation_accuracy_heatmap.png`; `validation_accuracy_heatmap.data.json`) supports sec. 4.1. It shows validation-only true accuracy and absolute recovery error for every evaluable model/persona/label term.
- `fig:evaluability-diagnostic` (`evaluability_diagnostic.png`; `evaluability_diagnostic.data.json`) supports sec. 4.2. It shows per-judge modal-vote fraction by model, with the 1.0 constant-classifier boundary.
- `fig:prompt-compliance` (`prompt_compliance.png`; `prompt_compliance.data.json`) supports sec. 4.2. It separates primary-valid cells, repaired-valid cells, final-invalid cells, and unavailable cells under the binary schema.
- `fig:temperature-schema-comparison` (`temperature_schema_comparison.png`; `temperature_schema_comparison.data.json`) supports sec. 4.2. It compares the old temperature-0.7 prompt baseline against the headline temperature-0.1 schema run on parseability and evaluability.
- `fig:scenario-hardness` (`scenario_hardness.png`; `scenario_hardness.data.json`) supports sec. 2.1. It plots the fraction of parsed calls that were bullish for each authored scenario, colored by the authored operational truth, with a 0.5 split line.
- `fig:recovery-mae-scorecard` (`recovery_mae_by_model.png`; `recovery_mae_by_model.data.json`) supports sec. 4.6. It shows recovery MAE per evaluable model against the 0.05 "excellent" and 0.102 sampling-noise reference lines.
- `fig:synthetic-convergence` (`synthetic_convergence.png`; `synthetic_convergence.data.json`) supports sec. 4.7. It plots recovery error against test size on log-log axes, with the `1/sqrt(Q)` reference and fitted slope -0.58.
- `fig:synthetic-alarm` (`synthetic_alarm.png`; `synthetic_alarm.data.json`) supports sec. 4.7. It shows alarm rate, discriminant behavior, and silent recovery error by error structure, with the `D = 0` alarm threshold.
- `fig:bootstrap-recovery` (`bootstrap_recovery_mae.png`; `bootstrap_recovery_mae.data.json`) supports sec. 4.1. It shows 2000 scenario-resampled recovery-MAE draws, with the point estimate, 95% CI, and 0.102 sampling-noise floor.
- `fig:synthetic-tiebreak` (`synthetic_tiebreak.png`; `synthetic_tiebreak.data.json`) supports sec. 4.8. It shows catastrophic mirror-selection and prevalence error as judge accuracy approaches random.
- `fig:synthetic-mv-vs-ei` (`synthetic_mv_vs_ei.png`; `synthetic_mv_vs_ei.data.json`) supports sec. 4.9. It compares EI and MV recovery MAE as judge accuracy falls.
- `fig:validation-matrix` (`validation_matrix.png`; `validation_matrix.data.json`) supports sec. 4.10. It reports generated checks for the manuscript's load-bearing statistical claims.

The figures are mutually reinforcing: fig. 1 defines the experimental boundary, fig. 9 states the thesis on the right axes, fig. 6 shows the per-judge mechanism that gates it, fig. 2 and fig. 3 show the positive recovery when the gate is cleared, fig. 4 explains what was actually recovered across models, and fig. 11 scores how close that recovery is to the 0.102 sampling-noise floor.

7 Supplemental material

Companion to the main manuscript. All numbers here are reproduced from generated artifacts: the cached LLM sweep in `outputs/llm/multi_llm_personas_evaluation/` (Ollama backend, temperature 0.1, N=3 personas, T=1 test, Q=64 scenarios, R=2 responses), the deterministic simulation/bootstrap outputs, and the offline validation matrix.

- S1 — The two-solution tie-break
- S1.5 — Worked example: the prevalence algebra on real votes
- S1.6 — Exactly when the imaginary-root alarm fires (and its blind spot)
- S2 — Full per-model label vote counts
- S3 — Full per-persona recovery (mistral:latest)
- S4 — Personas, scenarios, and authored truth
- S5 — Reproduction
- S6 — The evaluability diagnostic, defined
- S7 — Output file inventory

7.1 S1 — The two-solution tie-break

`ErrorIndependentEvaluation` (and `MajorityVotingEvaluation`) each return **two** algebraically consistent solutions for the trio's prevalence and per-persona accuracies. The two are a symmetry pair: the labelling we intend, and its *mirror image* in which every judge is treated as near-random/anti-correct and the prevalence is reflected about 0.5. Both satisfy the postulates exactly — the algebra alone cannot distinguish them.

We select with the standard answer-key-free assumption that **the judges are better than random**: keep the solution with the higher mean per-judge, per-label accuracy (`scripts/llm/_analysis.py::select_best`).

Two tempting alternatives are rejected on principle:

1. **Pick the solution closest to the true prevalence.** This peeks at the ground truth, so it is not unsupervised. It also fails outright when the two candidate prevalences are symmetric about 0.5 (e.g. 0.46 vs 0.54), where "closest to 0.5" is a coin flip.
2. **Pick by internal consistency.** Both solutions are *exactly* consistent — that is the whole point — so consistency cannot break the tie.

For mistral:latest, the selected solution had mean accuracy ≈ 0.89 against its mirror's ≈ 0.11; the tie-break is unambiguous. See the two solutions verbatim in `outputs/llm/multi_llm_personas_evaluation/models/mistral_latest/evaluation_error_independent.json`.

7.2 S1.5 — Worked example: the prevalence algebra on real votes

This section runs the sec. 3.6 formalism on **`mistral:latest`'s actual votes** — no toy numbers. Every value below is recomputed by `_values.py` from the on-disk `label_vote_counts.json` through `ntqr.r2`, so it cannot drift from the data.

Step 1 — the data sketch. From the trio's votes we read the all-bearish vote-pattern frequency and the frequency moments (label `b`):

Table 16: Real-vote frequency moments for `mistral:latest`, recomputed from `label_vote_counts.json`.

Quantity	Symbol	Value
All-`b` vote frequency	$f_{\beta\beta\beta}$	n/a
Trio frequency moment	m_{012}	n/a
Gap	$\Delta = f_{\beta\beta\beta} - m_{012}$	n/a
Pair moment (Opt,Neu)	$\delta_{(0,1)}$	n/a
Pair moment (Opt,Pes)	$\delta_{(0,2)}$	n/a
Pair moment (Neu,Pes)	$\delta_{(1,2)}$	n/a
Product of pair moments	$\prod_{(i,j)} \delta_{(i,j)}$	n/a

All three pair moments are **strictly positive** — every judge varies and the pairs co-vary — so the product does not vanish and the system is non-degenerate (contrast sec. 4.2, where a constant classifier drives a pair moment to zero).

Step 2 — the prevalence quadratic. With $a = \Delta^2 + 4\prod\delta$ and $c = \prod\delta$, the discriminant of $P_\alpha = \frac{1}{2} \mp \frac{1}{2}\sqrt{1 - 4c/a}$ evaluates to

$$1 - 4c/a = n/a \;>\; 0 \quad\Longrightarrow\quad \tfrac{1}{2}\sqrt{1 - 4c/a} = n/a. \tag{5}$$

The discriminant is **positive**, so the prevalence is real — the algebra finds no error-correlation violation (had it been negative, `ntqr` would raise `PrevalenceImaginaryException`; see sec. 3.6).

Step 3 — the two solutions. The $\mp$ yields the symmetry pair

$$P_\alpha \in \{\, n/a,\; n/a \,\}. \tag{6}$$

mirror images about $\frac{1}{2}$. The tie-break (sec. 7.1) selects **n/a** (higher mean accuracy), against the authored true prevalence of **n/a** — an absolute error of **n/a**, recovered with no answer key. Fixing this root then fixes all six per-persona, per-label accuracies via the linear maps of sec. 3.6 (tabulated in sec. 7.5).

7.3 S1.6 — Exactly when the imaginary-root alarm fires (and its blind spot)

The sec. 4.7 simulation finds, empirically, that the imaginary-prevalence alarm trips on an anti-correlated judge pair but **never** on positive common-mode correlation. That asymmetry is not a quirk of the draws — it falls straight out of the quadratic. Substituting $a = \Delta^2 + 4\prod\delta$ and $c = \prod\delta$ into the discriminant and simplifying,

$$D \;=\; 1 - \frac{4c}{a} \;=\; \frac{\Delta^2 + 4\prod\delta - 4\prod\delta}{\Delta^2 + 4\prod\delta} \;=\; \frac{\Delta^2}{\Delta^2 + 4\prod\delta}, \tag{7}$$

where $\Delta = f_{\beta\beta\beta} - m_{012}$ and $\prod\delta = \delta_{(0,1)}\,\delta_{(0,2)}\,\delta_{(1,2)}$ is the product of the three pairwise vote-moments. The numerator $\Delta^2 \geq 0$ is a square, so (for $\Delta \neq 0$) the alarm condition $D < 0$ is **exactly**

$$\Delta^2 + 4\prod\delta < 0 \quad\Longleftrightarrow\quad \prod\delta \;<\; -\tfrac{1}{4}\,\Delta^2 \;\leq\; 0. \tag{8}$$

Two consequences, both visible in sec. 4.7:

1. **The alarm is sufficient, never a false positive.** $D < 0$ forces $\prod\delta < 0$ — an *odd* number of the three pairs are anti-correlated in their votes. No genuinely error-independent test produces that in expectation, which is why the independent control fired in 0% of trials.
2. **It is blind to positive common-mode correlation.** If every pair co-varies positively ($\prod\delta > 0$), then $D \in (0, 1]$ for *any* strength of that correlation: the root stays real and the alarm is structurally silent. Shared errors only push every $\delta_{(i,j)}$ **more** positive, enlarging the denominator and driving $D \to 0^+$ without ever crossing it — precisely the sec. 4.7 curve where recovery degraded from 0.016 to 0.042 while the alarm never sounded. The real worked example above sits in this safe regime: all three $\delta_{(i,j)}$ are positive, so $D = n/a > 0$.

The practical reading: the imaginary-root alarm is a genuine, answer-key-free *sufficient* check, but a silent D is **not** a certificate of error-independence — only an absence of the specific anti-correlated signature it can see.

7.4 S2 — Full per-model label vote counts

Vote patterns are ordered `(Optimist, Neutral, Pessimist)`; `a` = bullish, `b` = bearish. The outer key is the **authored truth** of the scenario. Counts sum to 64 (40 bullish `a`, 24 bearish `b`).

`granite4.1:3b` — evaluable (every judge varies):

Truth	Vote pattern	Count
a (bullish)	a,a,a	33
a	a,b,b	3
a	a,a,b	3
a	b,b,b	1
b (bearish)	b,b,b	24

Per-judge: Optimist a×39/b×25, Neutral a×36/b×28, Pessimist a×33/b×31 → **no constant classifier**.

`mistral:latest` — evaluable (every judge varies):

Truth	Vote pattern	Count
a (bullish)	a,a,a	23
a	a,a,b	16
a	a,b,b	1
b (bearish)	b,b,b	19
b	a,b,b	5

Per-judge: Optimist a×45/b×19, Neutral a×39/b×25, Pessimist b×41/a×23 → **no constant classifier**.

`gemma2:2b` — evaluable (every judge varies):

Truth	Vote pattern	Count
a (bullish)	a,a,a	33
a	a,a,b	5
a	a,b,b	1
a	b,b,b	1
b (bearish)	b,b,b	24

Per-judge: Optimist a×39/b×25, Neutral a×38/b×26, Pessimist a×33/b×31 → **no constant classifier**.

`gemma3:4b` — evaluable (every judge varies):

Truth	Vote pattern	Count
a (bullish)	a,a,a	34
a	a,a,b	4
a	a,b,b	1
a	b,b,b	1
b (bearish)	b,b,b	19
b	a,b,b	5

Per-judge: Optimist a×44/b×20, Neutral a×38/b×26, Pessimist a×34/b×30 → **no constant classifier**.

`gemma4:latest` — evaluable (every judge varies):

Truth	Vote pattern	Count
a (bullish)	a,a,a	37
a	b,b,b	1
a	a,a,b	1
a	a,b,b	1
b (bearish)	b,b,b	23
b	a,a,b	1

Per-judge: Optimist a×40/b×24, Neutral a×39/b×25, Pessimist a×37/b×27 → **no constant classifier**.

`smollm2:135m-instruct-q4_K_S` — Optimist *and* Neutral *and* Pessimist constant:

Truth	Vote pattern	Count
a (bullish)	a,a,a	40
b (bearish)	a,a,a	24

Optimist, Neutral, Pessimist never varied → **two constant classifiers (Optimist, Neutral, Pessimist)**.

7.5 S3 — Full per-persona recovery (mistral:latest)

Unsupervised recovery (`ErrorIndependentEvaluation`, better-than-random branch) vs. supervised truth. Source: `models/mistral_latest/comparison.json` and `recovery_stats.json`.

Table 17: Full per-persona recovery table for `mistral:latest`, with the selected unsupervised branch compared to the supervised check.

Persona	Label	True	Recovered	Abs. error
Optimist	a	1.0000	1.0000	0.0000
Optimist	b	0.7917	0.7600	0.0317
Neutral	a	0.9750	1.0000	0.0250
Neutral	b	1.0000	1.0000	0.0000
Pessimist	a	0.5750	0.5897	0.0147
Pessimist	b	1.0000	1.0000	0.0000

- **Recovery MAE** = 0.0119 over six accuracy terms (close recovery, within the 0.102 sampling-noise floor).
- **Recovery MAE by label**: a = 0.0132, b = 0.0106.
- **Prevalence (label a)**: true 0.6250, recovered 0.6094, abs. error 0.0156.
- **Majority-voting recovery MAE** = 0.0119 (identical to error-independent here).

Exact fractions (from `evaluation_*.json`): recovered prevalence = 39/64; Optimist(b) = 19/25; Neutral = 1; Pessimist(a) = 23/39. The recovered values are close to the supervised truth, not exact; the residual error is the finite-test recovery error reported above and remains within the 0.102 sampling-noise floor.

7.6 S4 — Personas, scenarios, and authored truth

7.6.1 Persona system prompts (verbatim)

Prompt contract: `v3_binary_schema` (`4d157f84885a`).

Table 18: Versioned persona system prompts. The optimist and pessimist policies bias only genuinely mixed cases and explicitly allow evidence to override the persona.

Persona	System prompt
Optimist	You classify next-quarter equity direction from the stated facts only. Use the Optimist decision policy: in genuinely mixed cases, give slightly more weight to upside catalysts and recovery paths. If the evidence is exactly balanced, choose bullish. Do not force optimism when explicit negative evidence dominates; such cases should be bearish. Return valid JSON only.
Neutral	You classify next-quarter equity direction from the stated facts only. Use the Neutral decision policy: weigh positive and negative evidence symmetrically, with no directional prior. Make the call the evidence best supports; if it is close, still choose the stronger stated side. Return valid JSON only.
Pessimist	You classify next-quarter equity direction from the stated facts only. Use the Pessimist decision policy: in genuinely mixed cases, give slightly more weight to downside risks and execution failure. If the evidence is exactly balanced, choose bearish. Do not force pessimism when explicit positive evidence dominates; such cases should be bullish. Return valid JSON only.

Each persona receives the scenario and this user prompt:

Market scenario:[scenario text]Task: classify the asset's expected direction over the next quarter.Use only the facts stated above; do not assume outside market data.Choose exactly one allowed value for call: bullish or bearish.No neutral, no mixed, no abstention, and no explanation are allowed. If evidence is mixed, choose the allowed side implied by your persona policy.Return exactly one JSON object and no other text, for example {"call":"bullish"} or {"call":"bearish"}.

7.6.2 Scenarios and authored truth (Q = 64)

Table 19: Full generated scenario list and authored operational truth. These truths are validation-only and are not visible to the unsupervised evaluator.

ID	Truth	Scenario
s01	bullish	Revenue grew 42% YoY, beat estimates, and management raised full-year guidance.
s02	bearish	The company missed earnings for the third straight quarter and cut its dividend.
s03	bullish	A blockbuster drug just received FDA approval ahead of schedule with a broad label.
s04	bearish	Regulators opened a fraud probe and the CFO abruptly resigned this morning.
s05	bullish	Free cash flow doubled, the firm initiated a large buyback, and debt fell sharply.
s06	bearish	A key patent was invalidated and two generic competitors announced launches.
s07	bullish	The company signed a multi-year supply contract with the largest buyer in its sector.
s08	bearish	Inventory is piling up, margins compressed 600bps, and demand guidance was withdrawn.
s09	bullish	Same-store sales accelerated, traffic is up, and the loyalty program is growing fast.
s10	bearish	A major data breach exposed millions of customers and class-action suits are filed.
s11	bullish	The central bank signaled rate cuts and the company is highly rate-sensitive on the upside.
s12	bearish	The main product was recalled for a safety defect and shipments are halted.
s13	bullish	Order backlog hit a record and lead times extended, signaling robust demand.
s14	bearish	A larger rival slashed prices 30% to take share in the company's core market.
s15	bullish	Gross margins expanded for the fifth straight quarter as input costs fell.

ID	Truth	Scenario
s16	bearish	The firm issued a going-concern warning and is in breach of its loan covenants.
s17	bullish	A respected activist investor took a 9% stake and won three board seats.
s18	bearish	Key engineering talent is leaving and the flagship project slipped a full year.
s19	bullish	Earnings beat but guidance was merely in line; shares are near all-time highs. *(mixed)*
s20	bearish	Sales grew but margins thinned and the macro outlook for the sector is murky. *(mixed)*
s21	bullish	A turnaround plan is showing early traction though leverage remains elevated. *(mixed)*
s22	bearish	A one-time gain flattered profit while underlying organic growth stalled. *(mixed)*
s23	bullish	New management is cutting costs aggressively and reinvesting in a growing niche. *(mixed)*
s24	bearish	The stock is cheap on paper but sits in a structurally declining industry. *(mixed)*
s25	bullish	A new product line sold out within hours and the company doubled its production targets.
s26	bullish	The company won a multi-year government contract that triples its addressable market.
s27	bullish	Subscriber growth reaccelerated and monthly churn fell to a multi-year low.
s28	bullish	A long-running patent suit was settled in the company's favor with a large damages award.
s29	bullish	Operating leverage kicked in: revenue rose 20% while operating expenses stayed flat.
s30	bullish	The firm raised full-year guidance twice this quarter as bookings outpaced every forecast.
s31	bullish	A strategic partner took a minority stake at a premium and committed to joint development.
s32	bullish	Margins hit a record after the firm exited its lowest-margin segment and raised prices.
s33	bullish	Posted record annual revenue and free cash flow and initiated its first-ever dividend.
s34	bullish	A flagship product launch drew record preorders and sold out within a day.
s35	bullish	Regulators granted a breakthrough designation that fast-tracks the lead program.

ID	Truth	Scenario
s36	bullish	Closed an all-cash, immediately accretive acquisition that added no new debt.
s37	bullish	Its credit rating was upgraded to investment grade and borrowing costs fell sharply.
s38	bullish	Signed its largest-ever enterprise contract, a multi-year committed deal.
s39	bullish	Margins expanded as a new automated plant came online ahead of schedule.
s40	bullish	A major competitor exited the segment, leaving the firm the dominant supplier.
s41	bullish	Recurring revenue crossed 80% of sales as customer churn hit a record low.
s42	bullish	Won a landmark licensing deal that monetizes its previously idle patent portfolio.
s43	bullish	Order backlog doubled year over year on accelerating enterprise demand.
s44	bullish	Raised full-year guidance after a blowout quarter across every segment.
s45	bullish	A cornerstone investor took a large stake at a premium to the market price.
s46	bullish	Grew its net cash position while retiring high-coupon debt early.
s47	bullish	International expansion beat plan and the new region turned profitable quickly.
s48	bullish	A long-delayed product cleared its final certification with no conditions attached.
s49	bearish	Slashed full-year guidance twice within a single month as demand evaporated.
s50	bearish	A surprise goodwill writedown wiped out the entire quarter's profit.
s51	bearish	Its largest customer terminated a contract and switched to a direct rival.
s52	bearish	An accounting restatement triggered a delayed filing and an SEC inquiry.
s53	bearish	A safety recall halted production of its top-selling product line.
s54	bearish	Debt covenants were breached and lenders demanded immediate renegotiation.
s55	bearish	The CEO and CFO resigned in the same week amid a board dispute.
s56	bearish	A core patent expired and generic rivals immediately flooded the market.
s57	bullish	Revenue beat expectations but the firm guided conservatively citing macro caution.

ID	Truth	Scenario
s58	bearish	Margins held up but unit volumes quietly declined for a third straight quarter.
s59	bullish	A costly restructuring is underway but early customer cohorts show real traction.
s60	bearish	A buyback was announced even as several insiders sold into the news.
s61	bullish	Cash burn continues but a recent raise extends the runway by two years.
s62	bearish	Headline growth impressed but was driven entirely by one large one-off deal.
s63	bullish	The dividend was trimmed to fund a high-return capacity expansion.
s64	bearish	The stock screens cheap but sits in a slowly shrinking end market.

Authored prevalence: **40 bullish (a) / 24 bearish (b) → true prevalence of a = 0.625**. The deck is intentionally tilted off an even 50/50: an exactly balanced answer key would put the true prevalence at 1/2, a removable singularity of the error-independent evaluator where its accuracy equations degenerate to `NaN` (see sec. 3.3). The 24 clear bearish scenarios keep even the bullish-leaning personas casting both calls, so no judge collapses to a constant classifier.

7.7 S5 — Reproduction

All commands run from the repository root. The sweep is reproducible offline from the cached votes (`outputs/llm/.../cache.json`); only a fresh `--set refresh=true` run needs a live Ollama.

```
# 1. Project test suite (no network, no Ollama, no TeX):
.venv/bin/python -m pytest tests/ -q                        # 99 tests
#   or: cd python && ../.venv/bin/python -m pytest -q       # 22 library tests

# 2. Regenerate the multi-LLM sweep from the cache (JSON + figures):
.venv/bin/python scripts/llm/multi_llm_personas_evaluation.py

# 2b. Regenerate the deterministic synthetic-validation study (no network):
.venv/bin/python scripts/sim/synthetic_recovery_study.py

# 2c. Bootstrap a CI on the real recovery MAE (deterministic, offline):
.venv/bin/python scripts/sim/bootstrap_recovery_ci.py

# 2d. Regenerate the offline statistical validation matrix:
.venv/bin/python scripts/validation/statistical_validation_report.py

# 3. Re-inject the numbers into the manuscript (generated from templates):
.venv/bin/python scripts/manuscript/render.py
.venv/bin/python scripts/manuscript/render.py --check       # verify no drift
.venv/bin/python scripts/z_generate_manuscript_variables.py

# 4. Run the whole orchestrator harness including the LLM category:
.venv/bin/python scripts/run_all.py --include llm

# 5. Re-query live models (requires `ollama serve` and the pulled models):
.venv/bin/python scripts/llm/multi_llm_personas_evaluation.py \
    --set refresh=true --set timeout=45

# 6. Run a custom model list:
```

```
.venv/bin/python scripts/llm/multi_llm_personas_evaluation.py \
    --set models='["granite4.1:3b","mistral:latest"]'
```

Lint/format used in this study: `.venv/bin/ruff check scripts tests` and `.venv/bin/black --check -l 79 scripts tests`.

7.8 S6 — The evaluability diagnostic, defined

Computed from the vote matrix alone — no answer key, no solve (`scripts/llm/_analysis.py`):

- **`vote_variation(votes)`** → per judge: `distinct_calls`, `call_counts`, `modal_fraction` = (count of the most common call) / Q, and `is_constant` = (only one distinct call seen). `modal_fraction == 1.0` ⇔ `is_constant`.
- **`pairwise_disagreement(votes)`** → for each judge pair, the fraction of the Q scenarios on which they differ; plus the mean over the three pairs.
- **`evaluability_diagnostic(votes)`** → `predicted_evaluable` = (no judge `is_constant`), plus the named `constant_classifiers` and the mean pairwise disagreement.

Predictive check (this study). The orchestrator records, in `summary.json` → `evaluability_diagnostic`, whether the label-free `predicted_evaluable` set equals the set of models the ntqr solve actually evaluated. Result: `diagnostic_matched_outcome = true` — predicted evaluable = `gemma2:2b`, `gemma3:4b`, `gemma4:latest`, `granite4.1:3b`, `mistral:latest` = actually evaluated.

Disagreement vs. evaluability (the key contrast).

Model	Mean pairwise disagreement	Constant classifier?	Evaluable
smollm2:135m-instruct-q4_K_S	0.000	yes (Optimist, Neutral, Pessimist)	no
gemma4:latest	0.031	no	**yes**
granite4.1:3b	0.062	no	**yes**
gemma2:2b	0.062	no	**yes**
gemma3:4b	0.104	no	**yes**
mistral:latest	0.229	no	**yes**

In this low-temperature schema run, disagreement happens to sort the models because the only unevaluable model is a zero-disagreement constant-classifier collapse. That is not the NTQR solvability rule; it is a visible symptom of the per-judge rule. The diagnostic therefore keys on per-judge variation, not on ensemble disagreement.

7.9 S7 — Output file inventory

Per sweep, under `outputs/llm/multi_llm_personas_evaluation/`:

- `summary.json` — cross-model ranking, evaluability-diagnostic block, the requested/evaluated/failed model sets.
- `cache.json` — shared response cache, keyed by `backend / model / temperature / prompt-version / prompt-hash / persona / scenario`. New records include every raw attempt, parse method, schema-valid flag, invalid reason, parsed call, repair status, and prompt version/hash.
- `disagreement_vs_evaluability.png`, `recovery_mae_by_model.png`, `recovered_vs_true_scatter.png`, `evaluability_diagnostic.png`, `prompt_compliance.png`, `temperature_schema_comparison.png`, `scenario_hardness.png`, `best_model_recovery.png` — the figures, each with a `<stem>.data.json` sidecar.
- `models/<sanitized-model>/` — per model: `votes.json`, `label_vote_counts.json`, `evaluability_diagnostic.json`, `evaluation_error_independent.json`, `evaluation_majority_voting.json`, `evaluation_supervised.json`, `comparison.json`, `recovery_stats.json`, `comparison_majority_voting.json`, `recovery_stats_majority_voting.json`.

Degenerate models still emit `votes.json`, `label_vote_counts.json`, and `evaluability_diagnostic.json` (the inputs that explain the failure); they do not emit the recovery files, because there is no finite solution to record.

Additional generated artifacts:

- `outputs/sim/synthetic_recovery_study/` — deterministic convergence, dependence/alarm, tie-break, robustness, and EI-vs-MV sweeps; each embedded figure has a parseable `.data.json` sidecar.
- `outputs/sim/bootstrap_recovery_ci/` — scenario-bootstrap summary, `bootstrap_recovery_mae.png`, and `bootstrap_recovery_mae.data.json`.

- `outputs/validation/statistical_validation/` — offline validation report (`summary.json`), validation-matrix figure, and `validation_matrix.data.json`.

8 References

The bibliography is generated from `manuscript/references.bib`. The central package citation is the public NTQR 0.8 documentation [Corrada-Emmanuel et al., 2026], which documents the `ntqr.r2` evaluators and trio data sketches used here. This repository is cited as the reproducible experiment artifact [Friedman, 2026], and Ollama is cited for the local model runtime [Ollama, 2026a].

Supplemental material (tie-break derivation, full vote tables, per-persona recovery, scenario list, reproduction commands, diagnostic definition): **supplemental.md**.

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